Improving Semantic Segmentation in Aerial Imagery via Graph Reasoning and Disentangled Learning

Ruigang Niu, *Graduate Student Member, IEEE*, Xian Sun, *Senior Member, IEEE*, Yu Tian, Wenhui Diao, *Member, IEEE*, Yingchao Feng, *Graduate Student Member, IEEE*, and Kun Fu, *Member, IEEE*

***Abstract*— Semantic segmentation in aerial imagery is still an important, yet challenging task due to the complex characteristics of remote-sensing data. The critical issues consist of: 1) extreme foreground–background imbalance; 2) large intra-class variance; and 3) arbitrary-oriented, dense, and small objects. The above challenges make it unlikely to model the effective global interdependencies of semantic heterogeneous regions. Besides, general semantic segmentation methods suffer from feature ambiguity due to the joint feature learning paradigm, leading to inferior detail information. In this article, we propose an improved semantic segmentation framework to tackle these problems via graph reasoning (GR) and disentangled learning. On the one hand, a simple, yet effective GR unit is introduced to implement coordinate-interaction space mapping and perform relation reasoning over the graph. It can be deployed on the feature pyramid network (FPN) to exploit cross-stage multi-scale information. On the other hand, we propose a so-called disentangled learning paradigm to explicitly model the foreground and boundary objects, instantiated as foreground prior estimation (FPE) and boundary alignment (BA). The indication of the intermediate feature can be effectively emphasized to enhance the discriminative abilities of the network. Extensive experiments over iSAID, ISPRS Vaihingen, and the general Cityscapes datasets demonstrate the effectiveness and efficiency of the proposed framework over other state-of-the-art semantic segmentation methods.**

***Index Terms*— Aerial imagery, disentangled learning, feature alignment, graph convolutional networks (GCNs), semantic segmentation.**

Manuscript received September 22, 2021; accepted October 17, 2021. Date of publication October 19, 2021; date of current version February 8, 2022. This work was supported by the National Natural Science Foundation of China under Grant 61725105. *(Corresponding author: Xian Sun.)*

Ruigang Niu, Xian Sun, Yu Tian, Yingchao Feng, and Kun Fu are with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100190, China, also with the Key Laboratory of Network Information System Technology (NIST), Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100190, China, also with the University of Chinese Academy of Sciences, Beijing 100190, China, and also with the School of Electronic, Electrical and Communication Engineering, University of Chinese Academy of Sciences, Beijing 100190, China (e-mail: niuruigang18@mails.ucas.ac.cn; sunxian@aircas.ac.cn; tianyu181@mails.ucas.ac.cn; fengyingchao17@mails.ucas.edu.cn; kunfuiecas@gmail.com).

Wenhui Diao is with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100190, China, and also with the Key Laboratory of Network Information System Technology (NIST), Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100190, China (e-mail: diaowh@aircas.ac.cn).

Digital Object Identifier 10.1109/TGRS.2021.3121471

I. Introduction

As ONE of the most fundamental tasks in remote-sensing image interpretation, semantic segmentation has been widely applied to road extraction [1], infrastructure planning [2], urban change detection [3], and so on. It essentially aims at assigning class labels to each image pixel. Due to the unique imaging mechanism, scene characteristics, and ground sampling distance (GSD) of remote-sensing data, semantic segmentation in very-high-resolution (VHR) aerial imagery is still a challenging task.

Besides the common large-scale variation issue in natural scenes, the potential challenges of remote-sensing imagery are analyzed as follows (the examples are shown in Fig. 1).

1) *Foreground–background imbalance:* Foreground objects in aerial imagery are often overwhelmed by large ratio complex background. As illustrated in [4], the background ratio is more than 90% in some aerial datasets.
2) *Large intra-class variance, especially for background:* Objects of the same category often differ greatly in characteristics and scales. The background even contains all other categories that are not defined in the dataset, leading to high complexity.
3) *Arbitrary-oriented, dense, and small objects:* Aerial images are mainly captured by satellites or unmanned aerial vehicles (UAVs), so the objects can appear in various orientations and small objects are often densely arranged, such as small vehicles and planes.

Current semantic segmentation approaches are mainly based on the fully convolutional network (FCN) to aggregate multi-scale context. Specifically, DeepLab series [5]–[7] utilizes atrous spatial pyramid pooling (ASPP) module with different atrous rates to extract features in parallel and enlarge the receptive field to some extent. PSPNet [8] further adopts pyramid pooling module (PPM) to exploit global context with multi-scale features.

Recently, attention-based approaches [9]–[12] are proposed to explicitly capture long-range dependencies and further boost the segmentation performance in natural scenes. But they fail in modeling the effective global interdependencies of heterogeneous regions (such as boundaries where features change dramatically and backgrounds with larger intra-class variance)

1558-0644 © 2021 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:44:10 UTC from IEEE Xplore. Restrictions apply.

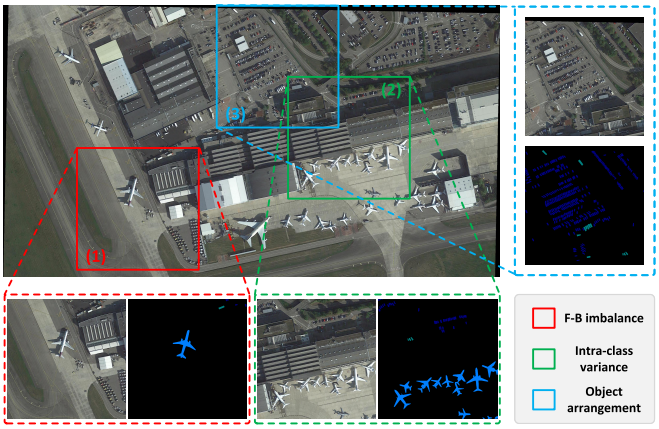

Fig. 1. Illustrations of potential challenges in remote-sensing aerial imagery. (1) Foreground–background imbalance (red dashed box). (2) Large intra-class variance. The background includes impervious surface, trees, buildings, and so on (green dashed box). (3) Arbitrary-oriented, dense, and small objects. Large and small vehicles are often densely arranged in various orientations (blue dashed box).

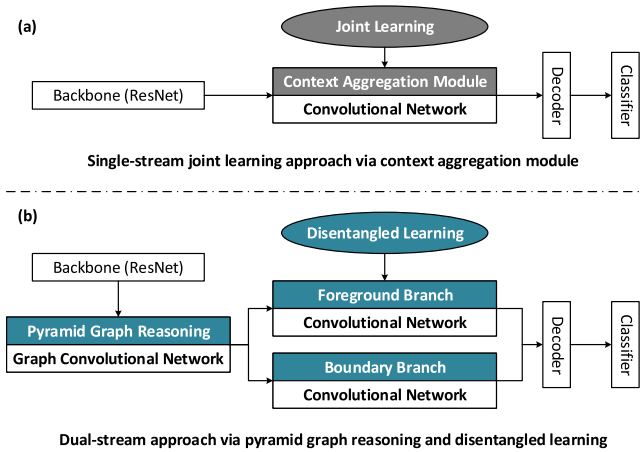

Fig. 2. Illustration of the segmentation pipeline. (a) In the previous works, the context aggregation modules (such as ASPP, PPM, Non-local, etc.) are directly appended on the top of the backbone. The corresponding feature representations are jointly learned indiscriminately through the single-stream network, leading to poor discriminative abilities. (b) In our method, we utilize the proposed PGR to model the global contexts. Then the encoded feature is fed into a decoupled dual-branch structure to explicitly learn the details of the foreground and boundary objects.

in such extreme foreground–background imbalance scenes. The attention mechanism forces the weighted aggregation of all positions to obtain the feature representation of each pixel, which brings prohibitive feature redundancy and invalid background information. Corresponding to the challenges 1) and 2) above, it generally leads to missing or false alarms issues due to the large ratio and high complexity of background. We summarize the above as the context modeling problem in extreme foreground–background imbalance scenarios.

To address the above issue, Zheng *et al.* [4] and Li *et al.* [13] enhance the perception of foreground objects via relation modules and sampling key points. The region shuffle attention in [14] alleviates the feature redundancy to a certain extent. Considering the object arrangement characteristics illustrated in challenge 3), small objects require more precise localization and feature fine-tuning. But the feature extraction of existing methods is indiscriminately coupled together, and the down-sampling operations lead to severe damage to the detailed information. As a result, foreground objects can be lost and boundary regions tend to be smooth and blurred, leading to inferior segmentation performance. It can be summarized as the feature ambiguity problem in semantic segmentation.

In this article, we propose an improved semantic segmentation framework to tackle the problems above via graph reasoning (GR) and disentangled learning.

For the context modeling problem, we consider using the graph convolutional network (GCN), which is well verified for information propagation [15]–[20]. Intuitively, the graph-structured data mainly relies on the generalized topology association of the data. Its inherent translation and rotation invariance avoids being too sensitive to the spatial dimension. However, there exist inherent semantic gaps between the graph and grid feature. Learning the graph representation from 2-D features and the edges that describe the graph are two key points. Motivated by the bilinear models [21] for fine-grained visual recognition, we devise an innovative GR unit to bridge the semantic gaps. We introduce hierarchical bilinear pooling (HBP) to capture high-order statistics with multi-scale representations and then project the input data from coordinate space to non-Euclidean interaction space through feature grouping. After simple, yet effective node reasoning over the graph (with learnable edges), the node features are re-projected to the original space. Note that the introduced HBP constructs graph-structured data from the perspective of channel dimension, thereby avoiding the extreme foreground–background imbalance issue in the spatial dimension. The entire GR unit is lightweight and can be implemented in a plug-and-play manner. Specifically, we apply it to the feature pyramid network (FPN) to exploit cross-stage multi-scale contextual information.

Toward the feature ambiguity problem, we argue that explicitly modeling foreground and boundary objects is crucial for semantic segmentation in aerial imagery. Inspired by the disentangled representation learning methods [22]–[24], we adopt the divide-and-conquer solution and introduce a so-called disentangled learning paradigm to perform foreground prior estimation (FPE) and boundary alignment (BA) within a decoupled dual branch. Concretely, we formulate the FPE into a Bayes' theorem manner and turn it into a supervised training process, estimating the foreground distribution through data prior. Besides, we design a BA module (BAM) to dynamically align boundary objects in an end-to-end trainable fashion. Experimental performance further indicates that the proposed disentangled learning paradigm is more conducive to the explicit feature extraction of the deep features, avoiding the ambiguity of coupled features in joint learning methods. The detailed comparisons of our method and the previous work are illustrated in Fig. 2.

We carry out comprehensive experiments on two challenging large-scale aerial datasets, that is, iSAID [25] and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:44:10 UTC from IEEE Xplore. Restrictions apply.

ISPRS Vaihingen [26]. We further verify the generalization of our method on the common Cityscapes [27] dataset. Our approach obtains superior performance compared with other state-of-the-art methods. In summary, the key contributions are threefold.

1) We analyze the potential challenges of aerial imagery and the corresponding issues in semantic segmentation and then propose a novel framework based on pyramid GR (PGR) and disentangled learning. It can effectively capture global interdependencies while emphasizing explicit feature representation.
2) We devise a simple, yet effective GR unit to model global context in extreme foreground–background imbalance scenarios. It can be cooperated with the FPN to exploit cross-stage multi-scale information.
3) A disentangled learning paradigm is introduced to explicitly model the foreground and boundary objects within a decoupled dual branch, that is, FPE and BA. It effectively avoids the ambiguity of coupled features in joint learning pipelines, thus preserving more precise detailed information for deep low-resolution features.

The remainder of this article is organized as follows. We first introduce related works in Section II and review some relevant theories in Section III. In Section IV, we elaborate our proposed method in detail. The ablation studies, comparisons with state-of-the-art, analysis, and visualization results are presented in Section V. Finally, the conclusion and future work are outlined in Section VI.

II. Related Works

In this section, we mainly focus on semantic segmentation in natural scenes and aerial imagery. Then, we introduce some GR approaches for semantic segmentation and the disentangled representation learning.

A. Semantic Segmentation in Natural Scenes

General semantic segmentation methods [5]–[8], [28]–[36] are based on FCN [37] for its effective semantic representation capabilities. They mainly focus on context modeling to obtain the large field of view of the network. To this end, Chen *et al.* [5] replace some down-sampling layers with dilated convolutions to enlarge the receptive field. Zhao *et al.* introduce a more straightforward region-based context aggregation through PPM. Recently, attention mechanism [38] is widely used in semantic segmentation to capture global context dependencies, instantiated as non-local operators [9]–[12], [39]–[45]. There are also several feature alignment methods [35], [46], [47] based on learnable interpolation strategy [48]–[50], which align adjacent features by learning transformation offsets. As Vision Transformer (ViT) [51] is gradually applied to vision tasks, relevant approaches [52], [53] tend to treat semantic segmentation as a sequence–sequence prediction task, replacing certain components of the convolutional network with transformers. These above methods have been well verified in natural scenes, but the performance is usually far from satisfactory if directly used in remote-sensing images. They ignore the complex characteristics and the main challenges (as illustrated in Section I) of remote-sensing data, especially the foreground–background imbalance issue. Some related improvements will be introduced in Section II-B.

B. Semantic Segmentation of Aerial Imagery

Semantic segmentation methods in remote-sensing imagery are generally inherited from the counterparts in natural scenes and optimized for specific application scenarios. In particular, several attention-based variants [14], [54]–[57] exploit more robust global contextual information through specially designed modules. Mi and Chen [58] introduce a superpixel-enhanced loss to refine boundary and object coherence. Peng *et al.* [59] propose a cross fusion network toward more accurate segmentation for small-scale objects. Feng *et al.* [60] utilize a boundary weighted loss to advance the segmentation performance for hard pixels. Further works focus on the specific modeling of the foreground objects. FarSeg [4] proposes a foreground-aware relation network to address the foreground–background imbalance and false alarm issues in VHR aerial imagery. PointFlow [13] introduces a point-wise affinity propagation module based on FPN to perform non-local operations over representative points. FactSeg [61] proposes a foreground activation-driven framework to address the small object segmentation problem. Unlike these methods, we propose a disentangled learning paradigm to explicitly model the foreground and boundary objects within a decoupled dual branch. It can effectively emphasize the indication of intermediate features, learn more accurate detailed information to promote the discriminability of the network.

C. Graph Reasoning

GCNs [62] have been widely used in various vision tasks [63]–[67] and proved to be an effective way of information propagation and relation reasoning. Here, we mainly focus on relevant methods for semantic segmentation. Concretely, GloRe [18] utilizes successive two-direction 1-D convolutions to simply implement relation reasoning over a fully connected graph. Both [15] and [16] aim at modeling relations between representative regions rather than individual pixels in the coordinate space. SpyGR [68] roughly performs the GR operation in the original feature space, ignoring the semantic gap between the coordinate space and the interaction space. CDGC [20] introduces a class-wise strategy to build graphs for each category and then perform graph-based reasoning in a coarse-to-fine manner. Zhang *et al.* [17] and Yu *et al.* [19] adaptively sample nodes in the graph and try to learn more compact representations for the fully connected graph. However, we argue that the sampling-based methods in spatial dimension cannot essentially avoid the problem of foreground–background imbalance. Hence, we design an HBP to construct graph-structured data from the perspective of channel dimension. It can effectively capture high-order statistics and exploit cross-stage multi-scale contexts cooperated with FPN.

D. Disentangled Representation Learning

Disentangled representation learning is designed to identify and disentangle the underlying explanatory factors of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:44:10 UTC from IEEE Xplore. Restrictions apply.

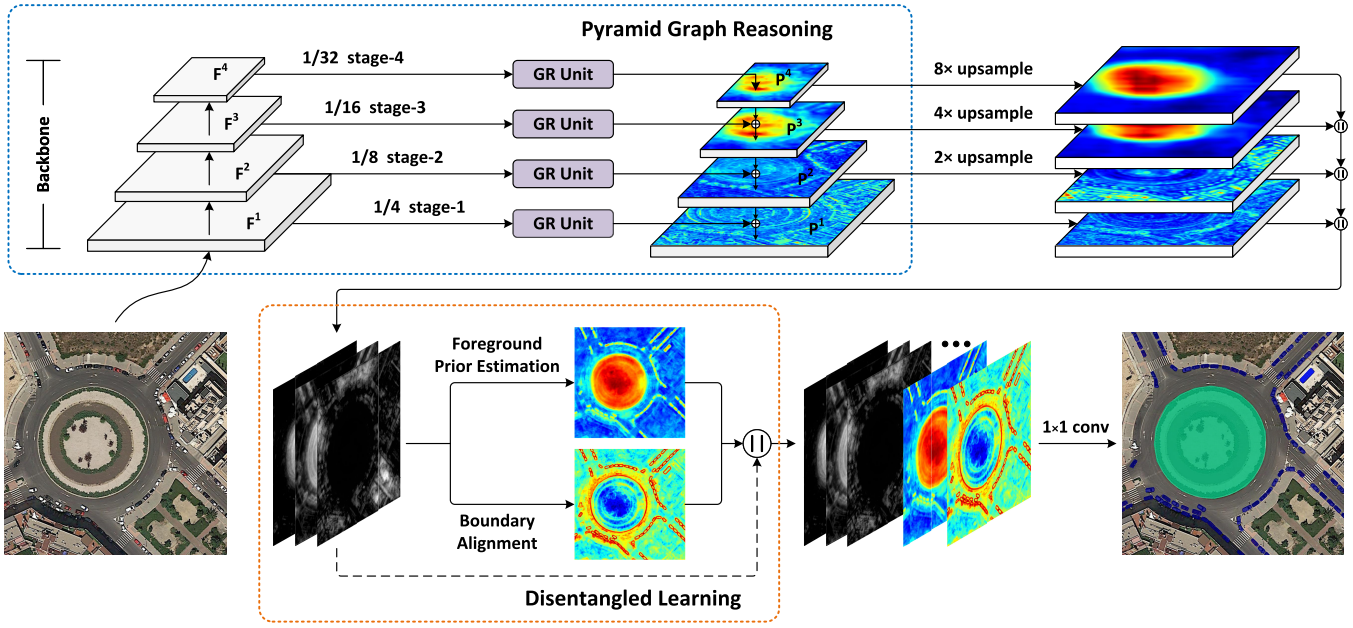

Fig. 3. Overall framework of our proposed method. The GR unit is embedded in each stage of the FPN. The decoupled dual branch structure is used to capture fine-grained foreground and boundary feature (best viewed in color). $\oplus$ represents element-wise summation and $\|$ denotes concatenation operation.

variations [22]. It has attracted considerable attention in terms of images representation learning [23], [69]–[71]. β-VAE [23] reformulates the standard variational autoencoder (VAE) [24], [72] framework as the constrained optimization problem with latent capacity constraint and independence prior pressures. Further approaches separate the information bottleneck term [73], [74] to achieve a better level of disentanglement [71], [75], [76]. In addition, Liu *et al.* [77] are the first to address the problem of representation disentanglement for cross-domain data. Ma *et al.* [78] introduce a disentangled VAE for learning disentangled representations from user behavior. DisenGCN [79] exploits the problem of disentangling the factors behind the formation of a graph and learns disentangled node representations. Intuitively, the disentangled representation learning can derive interpretable feature for describing the visual data, so that specific image attributes can be explicitly identified and manipulated. Therefore, we further focus on the disentangled representation at feature level and introduce a disentangled paradigm, explicitly modeling the foreground and boundary features, which are sensitive and crucial factors for remote sensing images.

III. Preliminaries

To better understand our proposed method, we first review two highly related theories, that is, the graph convolution and Bayes' theorem.

A. Graph Convolution

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ denotes a graph with N nodes, $A \in \mathbb{R}^{N\times N}$ (binary or weighted) is the adjacency matrix, and $D_{ii} = \sum_j A_{ij}$ represents the degree matrix. A basic GCN layer is defined by Kipf and Welling [62], and it can be formulated as

$$H^{(k+1)} = \sigma(\widetilde{A}H^{(k)}W^{(k)}) \tag{1}$$

where $\widetilde{A} = (D + I_N)^{-(1/2)}(A + I_N)(D + I_N)^{-(1/2)}$ is the normalized variant of the adjacency matrix (with self-loops) and $W^{(k)}$ is the layer-specific learnable weight matrix. I_N is the identity matrix and $\sigma(\cdot)$ denotes an activation function, such as $\tanh(\cdot)$ or $\mathrm{ReLU}(\cdot)$. $H^{(k)} \in \mathbb{R}^{N\times D}$ is the intermediate feature in the kth layer. The more detailed theoretical introduction is explained in Appendix A.

B. Bayes' Theorem

In probability theory and statistics, Bayes' theorem describes the probability of an event given prior knowledge of conditions that might be related to the event. Mathematically, Bayes' theorem is stated as the following equation:

$$P(Y \mid X) = \frac{P(X \mid Y)P(Y)}{P(X)} \tag{2}$$

where X and Y are different events and $P(X) \neq 0$. In Bayes' theorem, these notations have specific meanings when we consider that Y depends on its parameter X. $P(Y \mid X)$ is the posterior probability: the probability of Y occurring given parameter X. $P(X \mid Y)$ is the likelihood: the likelihood of X occurring given that Y is true. $P(Y)$ is the prior probability of Y, regardless of its parameter X. $P(X)$ is a standard constant, also known as the evidence factor. The naïve Bayes classifier is introduced in Appendix B.

IV. Method

In this section, we first present the overall pipeline of our proposed framework in Section IV-A. Then, we introduce our GR unit and disentangled learning paradigm in detail. Finally, we give some discussion on the choice of loss functions.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:44:10 UTC from IEEE Xplore. Restrictions apply.

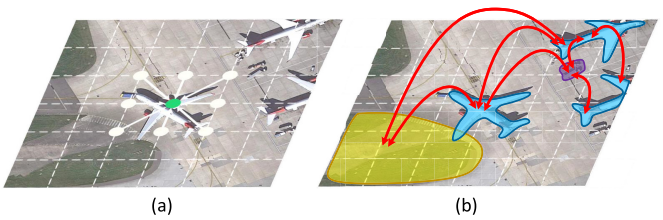

Fig. 4. Context modeling is crucial for semantic segmentation. Each spatial position of (a) standard convolution can be regarded as a node, with fixed neighbors and shared weights. The nodes in (b) graph-structured data can capture representations beyond regular grids, in which the adjacency matrix learns the dynamic interaction between nodes. (a) Local convolution. (b) Graph representation.

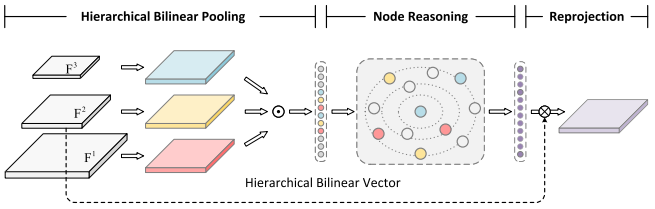

Fig. 5. Diagram of the proposed GR unit. We use HBP to map the input features from coordinate space to non-Euclidean interaction space. The node reasoning is implemented by two successive graph convolutional layers. Finally, we multiply the reasoned vector with the original input F^2 to reconstruct the feature representation.

A. Overall Framework

The overall pipeline of the proposed method is illustrated in Fig. 3, which is successively composed of basic network (ResNet-50/101), FPN with GR units, upsampling operators, cascaded dual branch structure, and the final classifier. We highlight the PGR and disentangled learning components with dashed boxes.

Specifically, stages 1–4 of the backbone network run in a bottom-up fashion, together with a top-down structure of the corresponding scale as the complete FPN framework. The input image is successively fed into the backbone, FPN with GR units, up-sampling, and concatenation operators and then generates a feature with 512 channels. After that, the disentangled learning paradigm is adopted for the encoded feature extraction, instantiated as a decoupled dual-branch structure to perform FPE and BA. Eventually, the output of the two branches are concatenated with the input feature and refined through a 1×1 conv to obtain the final segmentation map.

B. Pyramid Graph Reasoning

Toward the extreme foreground–background imbalance issue, the local convolution in the static grid feature suffers from redundant background modeling. We tend to build the more compact graph-structured representation and perform relation reasoning over the dynamic learnable graph. As illustrated in Fig. 4, local convolution is limited by the fixed neighboring nodes and shared weights, leading to the inferior modeling capacity. While the nodes in graph representation may refer to objects of different classes [20] or certain regions [18], which all depends on the specific design. Moreover, the interactions and weights between different nodes are dynamically learnable.

In this sections, we introduce a PGR structure to exploit multi-scale global context in a more effective manner. We first elaborate the GR unit and then discuss how to embed this meta-component into the FPN.

1) GR Unit: Dimension transformation and feature space mapping are the key points for grid-based feature maps to perform graph convolution. Thus, we are going to focus on graph projection, convolution, and reprojection operations. The diagram of the entire GR unit is shown in Fig. 5.

a) Graph projection: Simply put, given an intermediate feature map $F \in \mathbb{R}^{C\times H\times W}$ in coordinate space, we need to obtain 2-D graph-structured data $G \in \mathbb{R}^{N\times D}$. A straightforward way is to set $C = D$ as feature dimension and then flatten the spatial dimension to a 1-D vector with size of $H \times W$. However, it leads to prohibitive complexity if we define all pixels as graph nodes. To solve this issue, prior works [15], [16], [19] tend to sample or learn some representative pixels as the global descriptors. But the tradeoff between complexity and network capacity is tricky, and even excessive node sampling will lead to feature redundancy.

The more interesting methods [21], [80] adopt bilinear pooling to acquire second-order feature statistics and calculate global vectors. Nevertheless, they ignore the cross-stage multi-scale information in the basic network.

Inspired by the previous works [81], [82], we calculate the global vector via HBP operation. An example is illustrated in Fig. 5, for the output feature of stage-2 ($F^2 = [f_1^2, f_2^2, \ldots, f_{H_2W_2}^2] \in \mathbb{R}^{C\times H_2\times W_2}$), we consider the counterparts of its adjacent stages ($F^1 = [f_1^1, f_2^1, \ldots, f_{H_1W_1}^1] \in \mathbb{R}^{C\times H_1\times W_1}$ and $F^3 = [f_1^3, f_2^3, \ldots, f_{H_3W_3}^3] \in \mathbb{R}^{C\times H_3\times W_3}$) and adopt Hadamard product to aggregate features

$$G_{\text{proj}}(F^2, \mathcal{U}(F^1), \mathcal{U}(F^3)) = \frac{1}{H_2W_2}\sum_{i=1}^{H_2W_2} f_i^2 \circ f_i^{'1} \circ f_i^{'3} \quad (3)$$

where $\mathcal{U}(\cdot)$ is the bilinear interpolation to unify spatial dimension and $\circ$ means Hadamard product. f_i' represents the unified feature corresponding to f_i. Finally, the obtained hierarchical bilinear vector is divided into g groups to form the graph representation (as illustrated in Fig. 5). Each group can be regarded as a graph node with feature dimension d ($C = g \times d$).

b) Graph convolution.: We use two successive graph convolutional layers defined in (1) to further propagate information on the graph. Given the node feature $X \in \mathbb{R}^{N\times C}$, let $A_g \in \mathbb{R}^{N\times N}$ denote the adjacency matrix, and $W_g \in \mathbb{R}^{C\times F}$ is a trainable transformation matrix to control the dimension of the output node feature. A single graph convolution can be formulated as

$$Y = \sigma(A_g X W_g) \quad (4)$$

where $\sigma(\cdot)$ is the ReLU function and Y is the output feature.

As shown in Fig. 6, we design four forms of adjacency matrix: 1) fixed node connections, it allows one-hop neighbor and self-loop information propagation; 2) learnable fashion with identity matrix initialization (eye); 3) learnable fashion

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:44:10 UTC from IEEE Xplore. Restrictions apply.

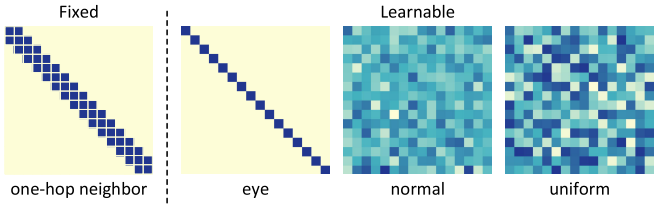

Fig. 6. Example of the adjacency matrix when $N = 16$.

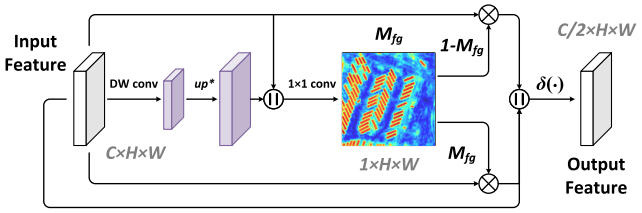

Fig. 7. Diagram of the proposed FPE. We perform matrix multiplication on the input feature with intra-prior (M_{fg}) and inter-prior ($1 - M_{\text{fg}}$) to exploit complementary context.

with normal distribution initialization; and 4) learnable fashion with uniform distribution initialization.

The detailed analysis of the design of the adjacency matrix and its applicability to remote-sensing scenarios are as follows. The adjacency matrix directly depicts the topology association of the graph. Empirically, we define the fixed undirected graph with self-loop and one-hop connection as the naive graph, which allows local interaction with weight 1. For highly representative (channel-wise statistics) graph nodes, the static graph is usually incapable. To this end, we further design the dynamic learnable graph with identity matrix initialization. It is more like a "residual connection" to maintain its initial behavior and facilitate the network optimization. Besides, the learnable fashions with normal and uniform distribution initialization are also considered for comparison. Note that to alleviate the extreme foreground–background imbalance problem in remote-sensing images, the HBP weakens the spatial dimension to generate channel-wise statistics, so the improved dynamic learnable graphs are preferable to ensure sufficient information interaction.

c) Graph reprojection.: The final step is to reshape the reasoned graph to global vector, and it can be regarded as the channel-wise statistic. So, we directly multiply it with the original input feature to recalibrate different channels following the previous work [83].

Specifically, we will carry out extensive ablation studies on each component of the GR unit in Section V-C5.

2) GR in FPN: As shown in Fig. 3, the input channel of F^1 to F^4 is 256, 512, 1024, and 2048, respectively. We first add a 1×1 conv to each lateral branch to normalize the channel dimension to 256 and then plug into a GR unit. Note that we only consider the unique adjacent stage feature for F^1 and F^4. Intuitively, the proposed PGR can effectively capture high-order feature statistics while taking into account the hierarchical multi-scale correlations.

C. Disentangled Learning

Current semantic segmentation methods mainly adopt joint learning methods without considering the ambiguity of coupled features. Besides the promising progress in the methods on representation disentanglement, recent works [39], [47], [84] also demonstrate that the decoupled design is more conducive to explicit feature extraction in various tasks. To this end, we propose a so-called disentangled learning paradigm, which can explicitly model foreground objects and align boundary features. The detailed workflow combined with loss functions is illustrated in Algorithm 1.

1) Foreground Prior Estimation: Prior work [85] has shown that the statistics of the input data are crucial for estimating the category distribution. Considering the extreme foreground–background imbalance, we intend to model the more important foreground objects through data prior. In particular, we formulate the FPE into a Bayes' theorem manner and turn it into a supervised training process.

Let $I \in \mathbb{R}^{512\times H\times W}$ denote the aggregated feature from the FPN framework. For simplicity, we define the class label set $\mathcal{Y} = \{c_{\text{fg}}, c_{\text{bg}}\}$, only considering the foreground c_{fg} and background c_{bg} categories. C is the random variable defined in $\mathcal{Y}$. For each spatial index j ($j \in [1, N], N = H \times W$), we can formulate the Bayes' theorem as

$$\frac{P(C_j = c_{\text{fg}})P(I_1 = i_1, \ldots, I_N = i_N \mid C_j = c_{\text{fg}})}{P(I_1 = i_1, \ldots, I_N = i_N)} = P(C_j = c_{\text{fg}} \mid I_1 = i_1, \ldots, I_N = i_N). \quad (5)$$

Bayes classifier tend to use the statistical information of the training samples to estimate the prior probability $P(C_j)$ and the likelihood $P(I_1 = i_1, \ldots, I_N = i_N \mid C_j)$. Then it calculates the product of two statistics for each class and outputs the category with the highest value. There exist two issues.

1) We can estimate the prior probability $P(C_j = c_{\text{fg}})$ by counting the number of foreground pixels for the entire dataset. But it is agnostic for a small number of samples of each random mini batch.
2) For continuous attributes, the likelihood generally obeys normal distribution, and it will lead to ambiguous meaning if modeled as a feature map.

To address the issues above, we explicitly learn the posterior probability $P(C_j = c_{\text{fg}} \mid I_1 = i_1, \ldots, I_N = i_N)$ supervised by the ground truth of foreground objects, which exactly meets the essential requirement of naïve Bayes (posterior probability maximization). It can be achieved via the corresponding foreground loss function, which will be elaborated in Section IV-D.

The diagram is shown in Fig. 7. In particular, we first obtain the compact feature through depthwise separable convolution and then the upsampled counterpart is concatenated with the original feature to generate the foreground mask $M_{\text{fg}} \in \mathbb{R}^{1\times H\times W}$ via 1×1 conv and sigmoid function. Based on the foreground mask, we can obtain the intra-prior (M_{fg}) and inter-prior ($1 - M_{\text{fg}}$). Finally, we capture the foreground and background context, which can be formulated as

$$B = \delta(M_{\text{fg}} \cdot I \parallel (1 - M_{\text{fg}}) \cdot I \parallel I) \quad (6)$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:44:10 UTC from IEEE Xplore. Restrictions apply.

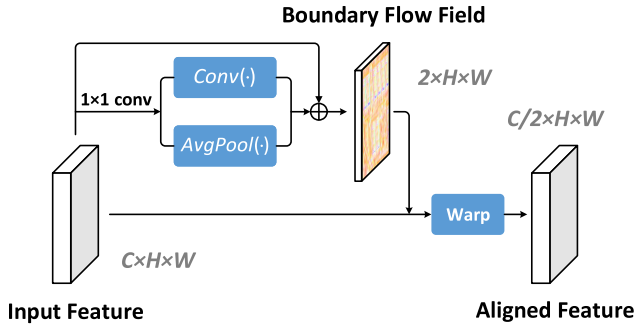

Fig. 8. Details of BAM.

where $\|$ denotes the concatenation operation and $B \in \mathbb{R}^{C/2\times H\times W}$ is the aggregated feature for further propagation. $\delta(\cdot)$ is the transformation function implemented by 3×3 conv $\rightarrow$ BN $\rightarrow$ ReLU.

2) Boundary Alignment: Limited by the object characteristics and image resolution, the boundary pixels are more error-prone in dense prediction tasks. From the perspective of the frequency domain, they correspond to the high-frequency parts of the original image, where the features change dramatically. Inspired by the feature warping operation in optical flow estimation [86], [87] task and related methods [47], [84] in segmentation, we design a BAM to align boundary features in an indirect supervised manner.

As shown in Fig. 8, given the same input feature $I \in \mathbb{R}^{C\times H\times W}$ obtained from FPN, we first feed it into a 1×1 conv to reduce the channel dimension, denoted by $I' \in \mathbb{R}^{C/2\times H\times W}$. Then we utilize both convolution-based and pooling-based methods to aggregate contextual information, which can provide rich semantic guidance for the following BA. Here, we introduce the idea of our boundary flow field, which models the similar feature representation as the flow field in optical flow. We regard it as the flow trend of boundary pixels in the feature alignment procedure. Simply put, it can be seen as an offset map, which rectifies the errors in the boundary area based on the adjacent positions. Concretely, the process of generating boundary flow field $\Delta_b \in \mathbb{R}^{H\times W\times 2}$ can be formulated as

$$\Delta_b = \gamma\left[I' \parallel \mathcal{U}(\text{Conv}(I') + \text{AvgPool}(I'))\right] \tag{7}$$

where $\mathcal{U}(\cdot)$ denotes the bilinear interpolation. Conv$(\cdot)$ is the 3×3 strided dilated convolution and AvgPool$(\cdot)$ is implemented by successive 3×3 strided average pooling and 1×1 conv. $\gamma(\cdot)$ is the 3×3 convolutional layer to predict the boundary flow field.

After obtaining the boundary flow field, each boundary representation j of the grid-based feature I' is then mapped into a new value k calculated by the learned flow field. The feature warping process is defined as follows:

$$\widehat{I'} = \sum_{j\in\mathcal{S}_b} \mathcal{W}\left(I'_j;\, \Delta_{b,j\rightarrow k}\right) \tag{8}$$

where $\widehat{I'}$ is the wrapped feature and $\mathcal{S}_b$ represents the set of boundary related pixels. $\mathcal{W}(x; w)$ denotes the results of warping the input x using the boundary flow field w.

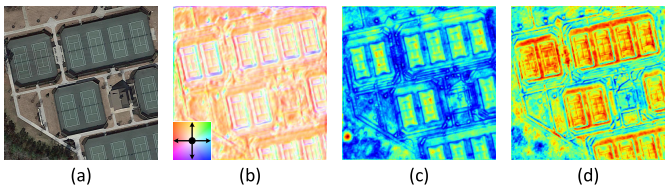

Fig. 9. Visualization of boundary flow field and feature maps in our BAM. (a) Input image. (b) Learned boundary flow field. As shown in the color wheel, flow direction is encoded with color and magnitude with color intensity (best viewed in color). (c) Feature before BAM. (d) Feature after BAM.

The aligned feature $\widehat{I'}$ concatenated with the original input are fed into a convolutional layer, obtaining the output of this branch.

Here, we discuss the set $\mathcal{S}_b$ of boundary related pixels. We first consider explicitly sampling the boundary pixels and then only perform the feature alignment on the counterparts. However, experiments show that the proportion of boundary pixels is too low, and the results of feature alignment of a few samples are not satisfactory. To tackle the issues, we adopt an adaptive BA strategy. The aligned feature is fed into a separate branch to learn the boundary map supervised by the binary ground truth. In this way, the warping process above tends to sample more points. $\mathcal{S}_b$ set will cover the boundaries as well as the associated neighbor pixels. The visualization of the learned boundary flow field and relevant feature representations are illustrated in Fig. 9.

D. Loss Functions

Considering the architecture design and disentangled learning paradigm, our network relies on richer supervisions to guide the learning the process. Accordingly, we employ four kinds of loss functions: segmentation loss, auxiliary loss, foreground loss, and boundary loss.

1) Segmentation Loss: Let $\mathcal{L}_{\text{ce}}$ denote the multi-class cross entropy loss

$$\mathcal{L}_{\text{ce}} = -\sum_{i=0}^{K-1} \mathbb{I}(y=i)\log(p_i) \tag{9}$$

where K is the number of categories. $\mathbb{I}(y=i)$ is an indicator factor, which is equal to 1 when the sample y belongs to the ith class, and 0 in other cases.

To alleviate the foreground–background imbalance problem, we optimize $\mathcal{L}_{\text{ce}}$ into the form of focal loss [88]. It compresses the relative value of the loss to a certain extent, so we further normalize it to the same magnitude. The segmentation loss can be expressed as

$$\mathcal{L}_{\text{seg}} = -\sum_{i=0}^{K-1} \mathbb{I}(y=i)\alpha(1-p_i)^{\gamma}\log(p_i) \tag{10}$$

where α is the normalization factor and γ is set to 2. The optimized segmentation loss forces the network to focus on the samples with lower confidence while maintaining the same magnitude.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:44:10 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 Decoupled Foreground and Boundary Supervision

Input:
$A \in \mathbb{R}^{512\times H\times W}$: the aggregated feature from FPN;
$\mathcal{T}_{train}$: training phase;
Output:
$\mathcal{W}_f$, $\mathcal{W}_b$: weight parameters of foreground prior estimation (FPE) and boundary alignment (BA);
while $\mathcal{T}_{train}$ **do**
  FPE:
  Generate the foreground mask M_{fg};
  Obtain the representation from FPE via Eq.6;
  BA:
  Generate the boundary flow field Δ_b;
  Align boundary features using Eq.8;
  Optimization:
  Calculate the foreground loss $\mathcal{L}_{fg}$ (Eq.14);
  Calculate the boundary loss $\mathcal{L}_{bd}$ (BCE);
  Calculate the overall loss $\mathcal{L}$ (Eq.15);
  Update the parameters $\mathcal{W}_f$ and $\mathcal{W}_b$:
    $\mathcal{W}_f, \mathcal{W}_b = \arg\min_{\{\mathcal{W}_f,\mathcal{W}_b\}} \mathcal{L}$;
end while
return $\mathcal{W}_f$, $\mathcal{W}_b$.

2) Auxiliary Loss: We follow the PSPNet [8] to add an auxiliary supervision after the third stage of the ResNet to improve the performance and accelerate the convergence of the model. The auxiliary loss $\mathcal{L}_{\text{aux}}$ is defined as the standard cross entropy loss with a weight of 0.4.

3) Foreground Loss: The foreground prior can be regarded as a binary classification task, in which case the cross entropy loss degrades into a binary cross entropy (BCE) loss. It can be formulated as

$$\mathcal{L}_{\text{bce}} = -[y_{\text{fg}} \log(p_{\text{fg}}) + (1 - y_{\text{fg}}) \log(1 - p_{\text{fg}})] \tag{11}$$

in which y_{fg} takes 1 when the sample belongs to the foreground objects; or else, it takes 0.

Apart from this, we also utilize a more straightforward dice loss [89] to measure the similarity between the prediction P and the ground truth G. The dice coefficient can be expressed as

$$\text{Dice} = \frac{2|P \cap G|}{|P| + |G|} \tag{12}$$

where $|P \cap G|$ is the intersection of P and G, where $|P|$ and $|G|$ represent the respective number of pixels. The numerator is multiplied by 2 to ensure that the interval is [0, 1]. Accordingly, we adopt a smooth dice loss as

$$\mathcal{L}_{\text{dice}} = 1 - \frac{2|P \cap G| + \epsilon}{|P| + |G| + \epsilon} \tag{13}$$

where ϵ is the smoothing coefficient to smooth the gradient and avoid some extreme situations. The total foreground loss is computed as

$$\mathcal{L}_{\text{fg}} = \mathcal{L}_{\text{bce}} + \lambda \mathcal{L}_{\text{dice}} \tag{14}$$

in which λ is set to 0.05. Note that dice loss is directly based on evaluation metrics, and the calculated gradient may be quite large, leading to unstable training. To this end, the choice $\lambda = 0.05$ is the optimal value obtained by the experiments (see Section V-C3).

4) Boundary Loss: The sparseness of boundary objects leads to the inapplicability of IoU-like loss. So, we simply use BCE loss as the boundary loss $\mathcal{L}_{\text{bd}}$. The specific definition is similar to (11).

Combining the aforementioned four loss functions, the overall loss can be formulated as

$$\mathcal{L} = \mathcal{L}_{\text{seg}} + \mathcal{L}_{\text{fg}} + \mathcal{L}_{\text{bd}} + \lambda_{\text{aux}} \mathcal{L}_{\text{aux}} \tag{15}$$

where λ_{aux} is empirically set to 0.4 as mentioned above.

Here, we give some analysis of the selection strategy and importance ranking of these loss functions. $\mathcal{L}_{\text{seg}}$ and $\mathcal{L}_{\text{aux}}$ are the essential basic loss functions of most existing semantic segmentation models, which are used to directly optimize the final segmentation results and need no further elaboration. The second important $\mathcal{L}_{\text{fg}}$ and $\mathcal{L}_{\text{bd}}$ are designed to ensure the representation learning of foreground and boundary features, respectively. Both foreground and boundary supervised learning are binary classification tasks in essence, so the BCE loss is preferable.

In addition, for the less-sparse foreground objects, we also consider the optimization directly based on evaluation metrics and select the dice loss, which is well verified in medical image segmentation. Note that the gradient of such IoU-lile loss often changes dramatically at the beginning of training phase and is more sensitive to the choice of the weight parameter λ. Therefore, we have carried out extensive tuning optimization and experiments have shown that it can further boost the performance under appropriate parameter selection.

V. Experiments

A. Datasets and Evaluation Metrics

1) iSAID: iSAID [25] is a recent large-scale segmentation benchmark for aerial imagery, which contains 2806 high-resolution images that range from 800 to 13 000 in width. The iSAID dataset consists of 655 451 annotated instances of 15 foreground categories. Note that we use the predefined 1411 for training and 458 images to evaluate our approach, for the annotations of the 937 images in the test set are unavailable. The mean of class-wise intersection over union (mIoU), mean F_1 score, and accuracy of foreground categories are adopted for evaluation.

2) Vaihingen: The Vaihingen [26] dataset consists of 33 VHR true orthophoto (TOP) tiles with the corresponding normalized digital surface model (DSM) data. Each tile has an average size of 2494 $\times$ 2064 and three spectral bands (near-infrared, red, green). The GSD of both TOP and DSM is 9 cm. We use the organizer defined 16 images (IDs: 1, 3, 5, 7, 11, 13, 15, 17, 21, 23, 26, 28, 30, 32, 34, 37) for training and the rest 17 images for testing. We use overall accuracy (OA), F_1 score, and mIoU for evaluation. Note that the OA is calculated over all categories including the background objects.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:44:10 UTC from IEEE Xplore. Restrictions apply.

3) Cityscapes: Cityscapes [27] dataset is a one of the most popular benchmarks for urban scene segmentation. It contains 5000 finely annotated images and 20 000 coarsely annotated images captured from 50 different cities. We only use the predefined finely annotated 2975 images for training and 500 images to evaluate our method. The standard mean interaction over union (mIoU) is reported for measuring the segmentation performance.

B. Implementation Details

1) Network Structure: We utilize ResNet-50/101 [90] pre-trained on ImageNet [91] as the backbone and remove the last fully connected layer. During the ablation studies, we choose vanilla ResNet-50+FPN as the baseline for explicit comparisons.

2) Training Settings: The network is end-to-end trained using stochastic gradient descent (SGD) with mini-batch 8. Note that the batch size is set to 4 for the ablation studies to speed up the experimental process, so the results are not optimal. We set the momentum and weight decay to 0.9 and 0.0001, respectively. The initial learning rate is set to 0.007 for iSAID dataset and 0.01 for Vaihingen and Cityscapes. We employ the poly-learning rate policy ($\text{lr} = \text{base_lr} * (1 - (\text{iter}/\text{max_ iter}))^{\text{power}}$) with power $= 0.9$. We crop the image into (896×896) patches for iSAID following the previous works [4], [13], (512×512) for Vaihingen and (769×769) for Cityscapes. All the experiments are conducted on 4× NVIDIA Tesla P100 GPU for 80k iterations. The standard BatchNorm is replaced with the InPlace-ABNSync [92] to synchronize the mean and standard deviation of BatchNorm across the four GPUs. During the training phase, we apply random crop, scaling (from 0.5 to 2.0), and horizontal flipping to augment the dataset.

C. Experiments on iSAID Dataset

1) Comparison With State of the Art: As shown in Fig. 10, we first visualize the size distribution of each category instance based on the instance annotations. The outliers also indicate large intra-class size variance. The intra-class variance is actually more about the difference in feature, and the learned representation for pixels of the same category may be quite different. These can be regarded as the hard examples. As illustrated at the bottom of Fig. 10, we give the visualization of feature distribution extracted by the backbone network. The subsequent experiments (see Section V-C7a) prove that the proposed PGR can effectively address this issue. In addition, we employ some complementary strategies to further boost the performance, including Online Hard Example Mining (OHEM) [17], Multi-Grid [6], and Multi-Scale inference [14].

We benchmark extensive results on iSAID validation set in Table I, most of which use ResNet-50 as the backbone. Note that OCR [43] employs much stronger backbone (HRNet-W48 [95]). Both SETR [53] and Swin-B [52] are based on Transformer [38], which is a more popular research field. It can be seen that attention-based methods do not work well in this dataset compared with the context aggregation approaches, such as [6] and [8]. FarSeg [4] introduce foreground-aware relation and optimization to gain better performance. In contrast, our approach achieves much more competitive results, that is, 65.1% mIoU on the validation set. And we further boost the performance to 66.8% mIoU with ResNet-101 backbone.

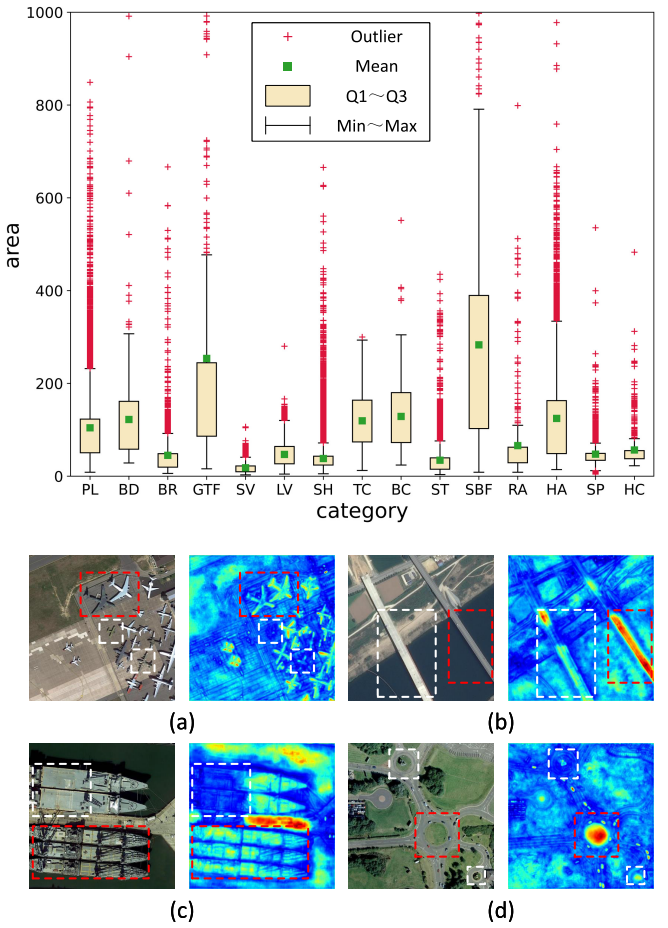

Fig. 10. Box plot shows the object size distribution of each category instance. The instance size is calculated by area $= (h \times w)^{1/2}$, where h and w are the height and width of the bounding box, respectively. The red crosses denote the outlier samples of a certain category. We further visualize some examples of the distributions of features in the same category. (a) Plane. (b) Bridge. (c) Ship. (d) Roundabout.

2) Ablation Study for the Proposed Approaches: We conduct ablation experiments to investigate the effect of different components of the proposed network, including the PGR, FPE, and BA. The vanilla ResNet-50 + FPN is adopted as the baseline. We add the GR unit on each lateral branch of the FPN and append the decoupled dual branch structure (FPE and BA) on the top of the baseline network.

As shown in Table II, the ResNet-50 + FPN baseline achieves a result of 59.4% mIoU. And adding the GR unit in FPN brings 1.8% mIoU improvements. The use of FPE and BA achieve 61.7% and 61.5% in mIoU, respectively. Furthermore, it outperforms the baseline by a large margin when we employ the integrated decoupled dual branches and achieves 63.2% mIoU. Finally, it yields the best 63.8% mIoU cooperated with the three components. The experimental results demonstrate that the graph representation for

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:44:10 UTC from IEEE Xplore. Restrictions apply.

TABLE I
RESULT COMPARISONS WITH STATE-OF-THE-ART ON iSAID VALIDATION SET

Method	Backbone	IoU per category(%)															mIoU(%)	mean F_1	Acc.(%)
		PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC			
EMANet [41]	ResNet-50	77.8	71.2	28.3	53.1	40.0	56.9	58.4	82.6	52.4	62.8	63.5	40.2	48.7	43.1	30.8	56.8	70.3	94.0
DenseASPP [31]	ResNet-50	78.2	66.9	28.4	53.5	38.2	57.4	60.8	82.4	53.7	50.1	69.9	54.8	51.2	43.4	25.7	57.1	70.6	94.2
Non-local [9]	ResNet-50	81.0	71.9	29.1	49.8	47.8	59.1	62.4	84.9	48.6	62.2	41.6	53.5	50.9	45.1	29.9	57.3	70.8	94.3
DANet [11]	ResNet-50	80.9	71.4	28.6	52.5	42.1	57.5	60.2	84.7	50.9	63.0	63.3	40.6	48.8	46.1	30.4	57.5	71.0	94.4
CCNet [10]	ResNet-50	78.8	72.4	30.5	52.2	44.4	53.1	56.3	85.3	49.2	63.2	67.1	51.8	48.7	43.6	29.2	57.8	71.3	94.6
Semantic FPN [93]	ResNet-50	81.2	71.5	33.8	52.2	45.4	60.1	63.5	87.1	57.8	61.5	60.2	59.0	51.5	46.6	31.2	60.1	73.5	94.8
DeepLab v3+ [7]	ResNet-50	82.9	71.4	32.2	52.0	48.5	60.1	64.5	81.1	55.7	68.5	61.0	60.2	53.0	47.1	30.8	60.5	73.9	94.8
PSPNet [8]	ResNet-50	82.8	76.8	31.2	53.4	47.3	58.2	63.9	86.5	56.7	68.3	67.4	52.6	53.0	44.0	33.3	60.9	74.2	95.1
SETR [53]	T-Large	81.5	74.5	30.8	52.3	48.8	58.6	63.4	86.9	57.4	67.9	68.0	61.6	52.9	46.2	32.6	61.4	74.4	95.3
UperNet [94]	Swin-B [52]	83.0	75.3	30.5	51.8	49.6	59.2	63.1	85.7	59.2	68.2	68.1	59.5	53.3	46.8	34.3	61.6	74.7	95.5
AlignSeg [35]	ResNet-50	82.9	76.2	28.7	52.0	50.3	60.7	67.4	86.2	62.1	68.9	71.2	56.2	**54.8**	45.7	31.2	62.1	75.0	95.6
OCR [43]	HRNet-W48	83.3	73.6	33.1	47.7	49.6	**61.4**	**67.8**	87.9	63.4	70.7	72.8	59.5	53.3	48.4	30.4	62.6	75.4	95.8
HMANet [14]	ResNet-50	83.8	74.7	29.0	54.6	50.3	59.7	65.4	**88.7**	60.5	**70.9**	70.2	62.9	51.9	51.4	32.6	62.6	75.5	95.8
SFNet [46]	ResNet-50	83.7	77.5	34.6	53.5	50.4	60.8	64.9	88.5	63.2	70.8	73.5	59.0	53.9	46.4	31.9	63.2	76.0	96.1
FarSeg [4]	ResNet-50	82.0	77.7	36.7	56.7	46.3	60.6	65.4	86.4	62.1	61.8	72.5	**71.4**	53.9	51.2	35.8	63.7	-	-
Ours	ResNet-50	**84.2**	**78.5**	**38.1**	**56.9**	**50.7**	60.5	65.5	**88.7**	**63.5**	69.2	**73.7**	68.7	52.8	**52.4**	**38.6**	**65.1**	**77.4**	**96.9**
Ours	ResNet-101	85.6	80.5	41.3	58.2	51.6	61.9	67.9	89.4	66.9	71.3	75.3	71.5	54.5	53.3	40.0	**66.8**	**79.1**	**97.5**

* The 15 foreground categories of iSAID data set: plane (PL), baseball diamond (BD), bridge (BR), ground track field (GTF), small vehicle (SV), large vehicle (LV), ship (SH), tennis court (TC), baseball court (BC), storage tank (ST), soccer ball field (SBF), roundabout (RA), harbor (HA), swimming pool (SP) and helicopter (HC).

TABLE II
ABLATION STUDY FOR EACH COMPONENT OF THE PROPOSED METHOD ON iSAID VALIDATION SET. PGR: PYRAMID GRAPH REASONING, FPE: FOREGROUND PRIOR ESTIMATION, BA: BOUNDARY ALIGNMENT

Method	PGR	FPE	BA	mIoU(%)
ResNet-50 + FPN				59.4
Ours	✓			61.2
		✓		61.7
			✓	61.5
		✓	✓	63.2
	✓	✓	✓	**63.8**

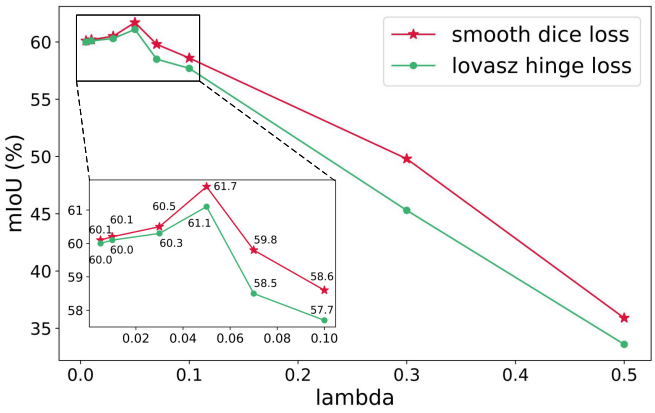

Fig. 11. Comparisons of different choices of weight parameter λ for smooth dice loss and lovasz hinge loss.

context modeling and the explicit extraction of foreground and boundary features are vital for object segmentation in such imbalanced scenes.

3) Ablation Study for the Foreground Loss and Weight Parameter λ: We use multiple loss functions to guide the learning procedure of the proposed network, among which the model is only more sensitive to the weight parameter λ of the smooth dice loss in (14). Specifically, the calculated gradient and loss values are quite large at the beginning of training, leading to unstable optimization. Note that we set $\mathcal{L}_{\text{seg}}$ in (15) to the standard cross entropy loss and remove the GR unit for comparison. We also tried the IoU-like Lovasz hinge loss [97] with the same weight parameter λ. The experimental results are shown in Fig. 11. It can be seen that the choice $\lambda = 0.05$ achieves the best result for the both loss functions. A large weight will lead to significant decrease in the experimental results, while the foreground loss will degenerate into the general BCE loss if the weight is much small. It is worth mentioning that we conduct three experiments for each choice of λ to eliminate the effects of training instability.

4) Ablation Study for the Loss Functions: We explore the importance of different kinds of supervisions and report the results in Table III. We can conclude that only with the supervised information, can the network effectively learn the expected feature representation and further improve the experimental results. Both foreground and boundary loss supervisions significantly improve the segmentation accuracy. It is further refined by the use of smooth dice loss. Note that the foreground loss follows the optimal parameter settings

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:44:10 UTC from IEEE Xplore. Restrictions apply.

TABLE III
ABLATION STUDY FOR THE LOSS FUNCTIONS

Method	$\mathcal{L}_{seg}$	$\mathcal{L}_{fg}$		$\mathcal{L}_{bd}$	$\mathcal{L}_{aux}$	mIoU(%)
		$\mathcal{L}_{bce}$	$\mathcal{L}_{dice}$			
Ours	✓					62.3
	✓				✓	62.5
	✓			✓	✓	62.9
	✓	✓		✓	✓	63.5
	✓	✓	✓	✓	✓	**63.8**

obtained in Section V-C3. Eventually, it yields the best result of 63.8% mIoU with the full loss functions.

5) Ablation Study for the GR Unit: For the proposed GR unit, we conduct ablation experiments from three perspectives: graph projection approaches, number of graph nodes, and design of the adjacency matrix.

a) Graph projection: We consider three graph projection operations from the channel dimension. The global average pooling (GAP) represents the squeeze (global information embedding) operation in SENet [83], which is commonly used for dimension reduction. The global hadamard pooling (GHP) introduced in [80] calculates the hadamard product for the input feature pairs. The introduced HBP can take into account the cross-stage multi-scale information while capturing the high-order statistics. As shown in Table IV (Proj.), the HBP achieves the best 63.1% mIoU with the same number of nodes and adjacency matrix.

b) Node.: In each lateral branch of the FPN, the channel dimension is unified to 256, which is exactly the length of the hierarchical bilinear vector. The number of graph nodes refers to the number of groups of the vector mentioned above. Here, we select a range of different values as shown in Table IV (Node), among which the choice node $=$ 16 yields better results than others. Empirically, an identical number of nodes for different feature dimensions may not always be optimal, and this is probably because 16 is a moderate value for 256 channels. Som further performance improvement may be achieved by turning more adapted hyper-parameters.

c) Adjacency matrix.: The adjacency matrix indicates the information interaction relationship between different nodes. We conduct ablation studies on the adjacency matrices illustrated in Fig. 6, and the experimental results are detailed in Table IV (Adj.). We can conclude that the learnable method is better compared with the fixed one-hop neighbor and self-loop connection. Finally, it achieves a result of 63.8% mIoU when we choose HBP, node $=$ 16, and the learnable adjacency matrix with uniform initialization.

6) Comparison With Context Modules: We further compare the performance and efficiency with some well verified context aggregation methods, attention-based methods, and graph-based methods, including ASPP [7], PPM [8], SA [9], CCA [10], OCR [43], GAT [96], and GloRe [18]. For a fair comparison, we only plug each module into the top lateral branch (stage-4) of the FPN architecture. The results are detailed in Table V. Both "+ASPP" and "+PPM" achieve better performance than other attention-based and graph-based modules in such complicated scenes, but they bring much more extra parameters and computation costs [floating point operations (FLOPs)]. Attention-based methods have been widely validated for natural scenes, but usually do not work well in remote-sensing scenarios. In general, graph-based methods seldom use convolution operations, leading to fewer parameters and less computation. Note that the graph attention network, denoted as "+GAT," utilizes pure attention mechanism without parameters. In contrast, our GR unit achieves the best trade-off between performance and computation cost. It can be observed that the proposed "+GR Unit" improve the baseline by 1.5% mIoU with negligible parameters and FLOPs. The mIoU is further boosted to 61.2% if GR Units are added to each lateral branch of FPN, and it will not bring too much computation cost.

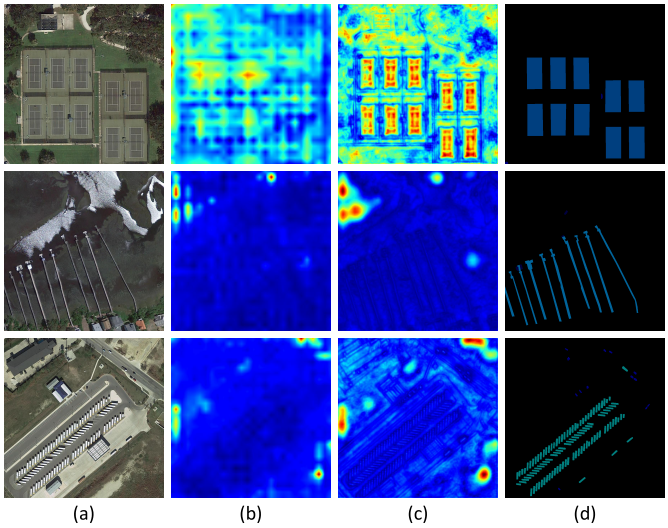

Fig. 12. Visualization of the features before and after PGR on the iSAID validation set. (a) Input image. (b) Feature w/o PGR (after the stage-4 of ResNet). (c) Feature w/ PGR. (d) Ground truth.

7) Visualization Analysis:

a) Pyramid graph reasoning: The effect of the PGR can be visualized in Fig. 12. For each image, we visualize the feature after the backbone and our PGR architecture, that is, Fig. 12(b) and (c). It can be seen that the proposed PGR does not change the general distribution of the input feature, but provides more detailed contextual information. There is a high consistency between objects of the same category, which exactly solves the drawbacks of the attention mechanism in such extreme scenarios.

b) Learned foreground and boundary map.: To better understand the effect of our decoupled dual-branch structure, we visualize the learned foreground and boundary maps in Fig. 13(b) and (c), which explicitly extract the body and edge parts of the foreground objects, respectively. Fig. 13(d) denotes the corresponding boundary flow field in the BAM. Note that the visualization code in [87] is adopted in the procedure. Fig. 13(e) represents the ground truth of the boundary pixels. Considering the issue of dense small objects, the boundary width is set to one pixel during the training phase.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:44:10 UTC from IEEE Xplore. Restrictions apply.

TABLE IV

ABLATION STUDY FOR EACH COMPONENT OF THE PROPOSED GR UNIT, INCLUDING THE GRAPH PROJECTION OPERATION, NUMBER OF GRAPH NODES, AND DESIGN OF THE ADJACENCY MATRIX. WE CHOOSE VANILLA RESNET-50 + FPN AS THE BASELINE

Method	Graph Projection			Node	Adjacency Matrix		mIoU(%)
	GAP	GHP	HBP		Fixed	Learnable	
Baseline	-	-	-	-	-	-	59.4
+ GR Unit (Proj.)	✓			16	one-hop		62.3
		✓		16	one-hop		62.5
			✓	16	one-hop		63.1
+ GR Unit (Node)			✓	4	one-hop		61.9
			✓	8	one-hop		62.7
			✓	16	one-hop		63.1
			✓	32	one-hop		62.8
			✓	64	one-hop		62.3
+ GR Unit (Adj.)			✓	16	one-hop		63.1
			✓	16		eye init	63.6
			✓	16		normal init	63.5
			✓	**16**		**uniform init**	**63.8**

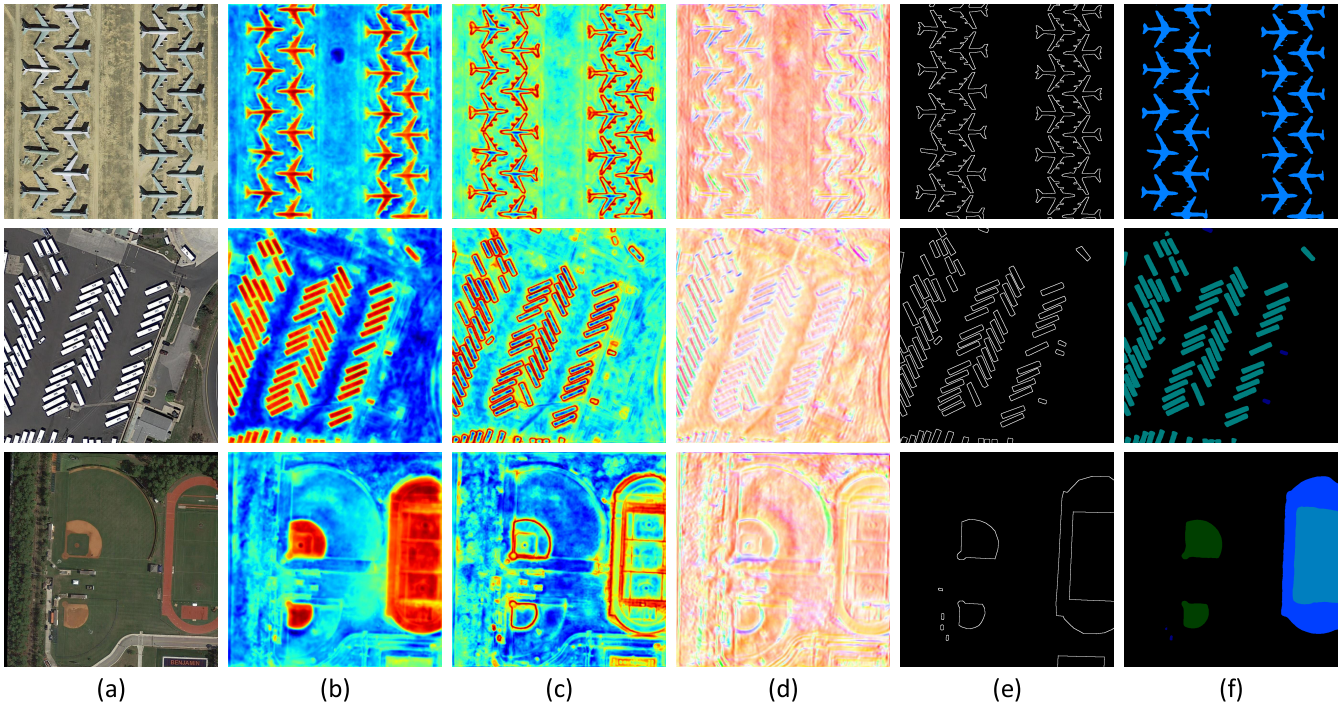

Fig. 13. Visualization of the learned foreground and boundary map and the corresponding flow field on the iSAID validation set. (a) Input image. (b) Foreground map. (c) Boundary map. (d) Boundary flow field. (e) Boundary ground truth. (f) Segmentation result. Best viewed in color.

8) Qualitative Results: We visualize some 896 × 896 patches between our method and the baseline (ResNet-50 + FPN) in Fig. 14. Subjectively, the segmentation effect is improved significantly, which is embodied in the discriminant ability between foreground objects and complex backgrounds. There is basically no missing objects and false alarms issues, which demonstrates the superiority of FPE and modeling global context in non-Euclidean interaction space. Besides, the proposed BAM also helps to refine the detailed information of the boundary regions and maintain the internal consistency of the foreground objects, especially for dense scenes.

D. Experiments on Vaihingen Dataset

To demonstrate the effectiveness of our proposed method, we carry out experiments on the more commonly used ISPRS Vaihingen benchmark. Note that there are two kinds of ground truth for this dataset: the reference with and without eroded boundaries. Taking into account the importance of boundary

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:44:10 UTC from IEEE Xplore. Restrictions apply.

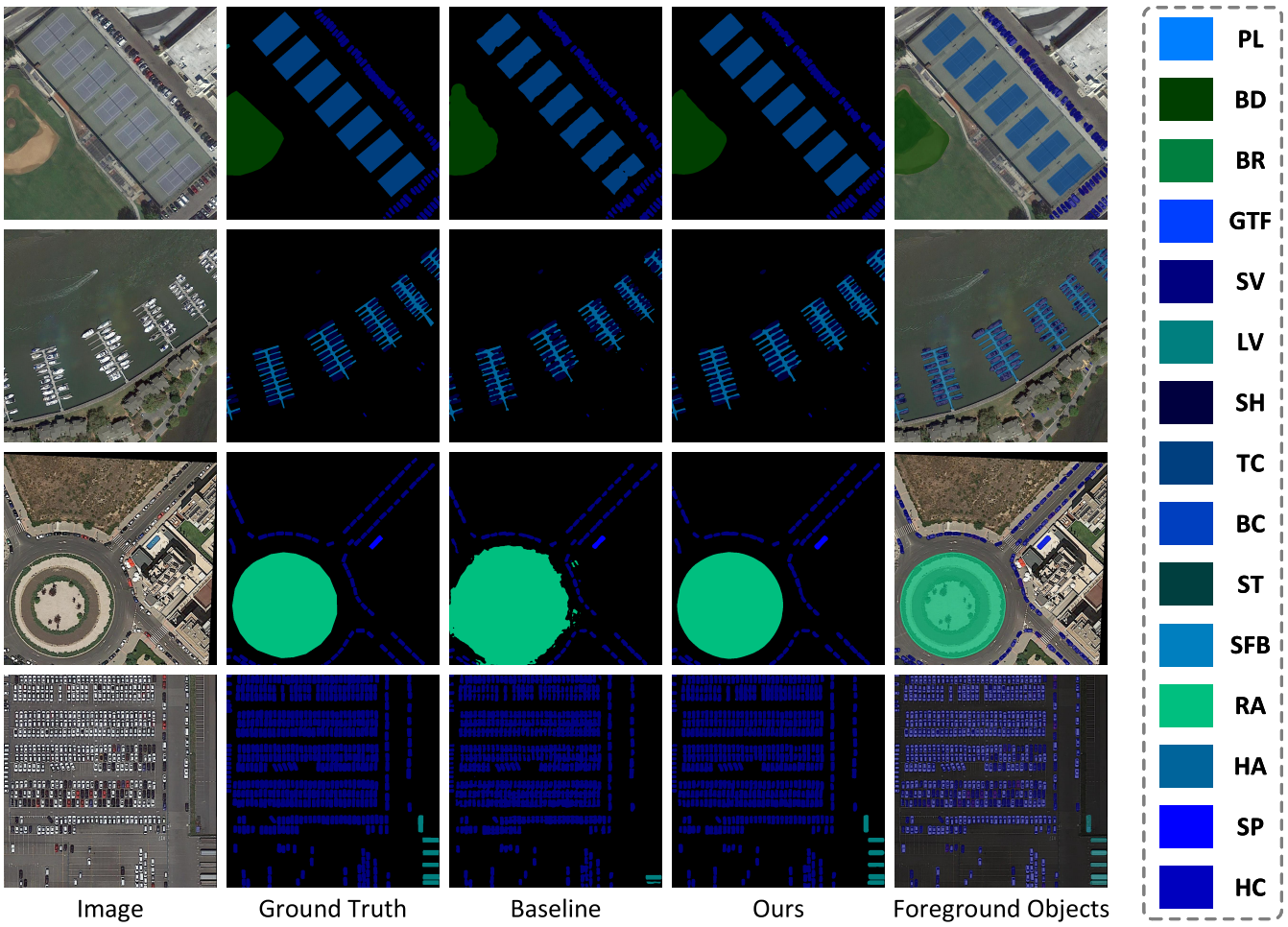

Fig. 14. Qualitative comparisons between our method and the baseline (ResNet-101 + FPN) on the iSAID validation set.

TABLE V

COMPARISON WITH CONTEXT AGGREGATION, ATTENTION-BASED, AND GRAPH-BASED APPROACHES. THE PARAMETER AND GFLOPS ARE CALCULATED WITH THE INPUT SIZE 896 × 896. THUS, THE SIZE OF CORRESPONDING FEATURE FED INTO EACH MODULE IS 1 × 2048 × 28 × 28

Method	Params	FLOPs	mIoU(%)
ResNet-50 + FPN	30.2M	345.3G	59.4
+ ASPP [7]	+14.8M	+11.0G	60.5
+ PPM [8]	+8.8M	+5.2G	60.9
+ SA [9]	+3.5M	+3.5G	59.2
+ CCA [10]	+5.0M	+3.8G	59.4
+ OCR [43]	+3.1M	+1.6G	59.7
+ GAT [96]	-	-	60.2
+ GloRe [18]	+2.8M	+2.0G	60.4
+ GR Unit (Ours)	**+0.5M**	**+0.4G**	**60.9**
+ 4×GR Unit (Ours)	**+0.9M**	**+2.7G**	**61.2**

objects, we also use the ground truth without eroded boundaries. The numerical comparisons with other state-of-the-art approaches are detailed in Table VI. For common quality measures (w/ eroded boundary), our method yields the best results of 91.8% in OA and 84.2% in mIoU. Compared with the well verified attention-based methods, our context modeling method is more effective and the disentangled feature representation is more interpretable. Especially for small car objects, our method outperforms other approaches by a large margin, achieving 90.1% F_1 score. Experiments show that the OA of the same approaches will be reduced by about 3% using the full reference without eroded boundaries, such as DeepLab v3+ [7] and PSPNet [8]. In contrast, the OA of our method is only reduced by 2.5% under the same experimental settings, which proves the effectiveness of the BAM.

Qualitative results are presented in Fig. 15. We choose ResNet-101 + FPN as the baseline. It can be seen that our method can provide more accurate predictions for the foreground objects and boundary regions. For small car objects, there is almost no missing or false alarm issue. It remarkably boosts the performance especially for the indistinguishable low vegetation and tree, which proves the effectiveness of graph representation modeling.

E. Experiments on Cityscapes Dataset

We further conduct experiments on the common Cityscapes dataset to verify the generalization of our approach. For a fair

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:44:10 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
COMPARISONS WITH THE STATE-OF-THE-ART METHODS ON VAIHINGEN TEST SET

Method	Backbone	F_1 per category(%)					mean F_1	OA(%)	mIoU(%)
		Imp. surf.	Building	Low veg.	Tree	Car			
W/O Eroded Boundary (Full Reference)									
U-Net [98]	-	-	-	-	-	-	81.0	84.0	68.7
DenseASPP [31]	ResNet-101	88.2	91.5	79.7	84.3	78.5	84.4	86.9	73.8
UperNet [94]	Swin-B [52]	88.8	92.0	80.2	85.6	78.3	85.0	87.2	75.2
DeepLab v3+ [7]	ResNet-101	89.5	92.2	80.8	86.5	77.6	85.3	87.7	76.0
AlignSeg [35]	ResNet-101	89.8	92.6	81.2	86.5	79.4	85.9	87.9	76.5
PSPNet [8]	ResNet-101	90.2	93.8	80.9	86.7	81.7	86.7	88.0	76.8
CCNet [10]	ResNet-101	90.3	93.6	81.5	86.5	82.0	86.8	88.1	76.9
OCR [43]	HRNet-W48	90.5	93.8	81.8	87.1	82.2	87.1	88.4	77.3
Ours	ResNet-101	**90.8**	**94.2**	**82.5**	**87.8**	**83.5**	**87.8**	**89.3**	**78.7**
W/ Eroded Boundary									
FCN [37]	VGG-16	88.7	92.8	76.3	86.7	74.2	83.7	86.5	72.7
RoteEqNet [99]	-	89.5	94.8	77.5	86.5	72.6	84.2	87.5	-
S-RA-FCN [54]	VGG-16	91.5	95.0	80.6	88.6	87.1	88.6	89.2	79.8
V-FuseNet [100]	-	92.0	94.4	84.5	89.9	86.3	89.4	90.0	-
DLR_9 [101]	-	92.4	95.2	83.9	89.9	81.2	88.5	90.3	-
DeepLab v3+ [7]	ResNet-101	92.4	95.2	84.3	89.5	86.5	89.6	90.6	81.5
Semantic FPN [93]	ResNet-101	92.6	95.1	84.5	89.5	86.6	89.7	90.7	81.8
PSPNet [8]	ResNet-101	92.8	95.5	84.5	90.0	88.6	90.3	90.9	82.6
CASIA2 [102]	ResNet-101	93.2	96.0	84.7	89.9	86.7	90.1	91.1	-
HMANet [14]	ResNet-101	93.5	95.9	85.4	90.4	89.6	91.0	91.4	83.5
HUSTW5 [103]	ResNet-101	93.3	**96.1**	86.4	**90.8**	74.6	88.2	91.6	-
Ours	ResNet-101	**93.7**	95.8	**86.5**	**90.8**	**90.1**	**91.4**	**91.8**	**84.2**

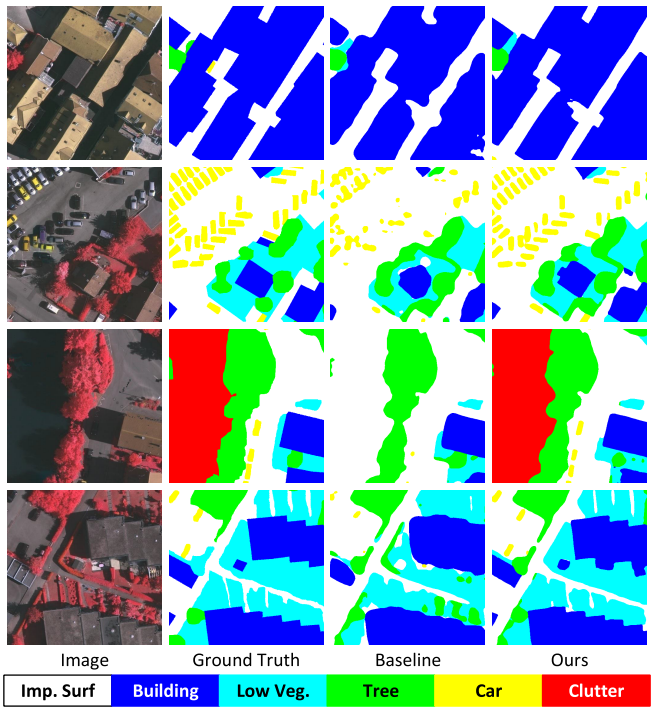

Fig. 15. Visualization comparisons between our method and the baseline (ResNet-101 + FPN) on the ISPRS Vaihingen test set.

comparison, we use ResNet-50 as the backbone for all models and choose the vanilla ResNet-50 + FPN as the baseline. All models are trained using only finely annotated 2975 images and validated on the predefined validation set (500 images). Note that for the Cityscapes dataset, there is no specifically defined background category, so we set the unlabeled and ignored pixels as the background.

TABLE VII
PERFORMANCE COMPARISONS ON CITYSCAPES VALIDATION SET

Method	Backbone	mIoU(%)
RefineNet [30]	ResNet-50	73.2
PSANet [104]	ResNet-50	76.1
PSPNet [8]	ResNet-50	76.8
DeepLab v3+ [7]	ResNet-50	77.9
Non-local [9]	ResNet-50	78.1
GloRe [18]	ResNet-50	78.3
CPNet [105]	ResNet-50	78.4
CCNet [10]	ResNet-50	78.5
DANet [11]	ResNet-50	78.6
GCNet [42]	ResNet-50	78.6
AlignSeg [35]	ResNet-50	78.8
OCR [43]	ResNet-50	78.9
ResNet-50 + FPN	ResNet-50	72.8
Ours	ResNet-50	**79.5**

The numerical results on the validation set are shown in Table VII. It can be seen that our approach achieves the best 79.5% mIoU, which outperforms the baseline by a large margin. It also surpasses the well-verified

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:44:10 UTC from IEEE Xplore. Restrictions apply.

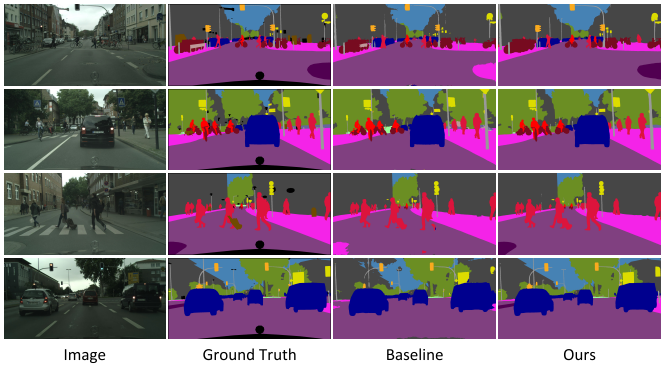

Fig. 16. Qualitative comparisons between our approach and the baseline (ResNet-50 + FPN) on the Cityscapes validation set.

context-based methods (GloRe [18], CPNet [105], AlignSeg [35], etc.), and attention-based methods (Non-local [9], CCNet [10], GCNet [42], OCR [43], etc.).

The qualitative results are illustrated in Fig. 16. Compared with the baseline, our model has a better perception of foreground objects and can predict precise and thin boundary features. It also maintains a higher consistency within the objects, which demonstrate the effectiveness of modeling global context via graph representation.

VI. Conclusion and Future Work

In this article, we focus on the context modeling issue in extreme foreground–background imbalance scenes and feature ambiguity problem for semantic segmentation in aerial imagery. Thus, we propose an improved semantic segmentation framework that contains two primary components: PGR and disentangled learning paradigm. In particular, the feature representations of coordinate space are projected into non-Euclidean interaction space for relation reasoning via the proposed GR unit. It can be cooperated with the FPN to exploit cross-stage multi-scale contextual information. The disentangled learning paradigm is manifested as explicitly modeling foreground and boundary objects. We formulate the proposed FPE into a Bayes' theorem manner and estimate foreground distribution through data prior. Besides, a simple, yet effective BAM is introduced to align boundary objects in an end-to-end trainable fashion. Extensive experiments on the iSAID, ISPRS Vaihingen benchmark, and Cityscapes dataset demonstrate the superior performance of the proposed method.

Experimental results indicate that the proposed method can be effectively applied to dense prediction tasks. In the future, we consider treating such dense prediction tasks as sequence tasks and explore sequential models such as ViT.

Appendix A
Graph Convolutional Networks

As the basic unit of GCN, graph convolution can be defined as the convolutional operation on graph-structured data in non-Euclidean space. Like deep convolutional neural networks, the key insight of the GCN method is that we can build very powerful networks by stacking very simple graph convolutional layers. The aforementioned GCN layer is defined as

$$H^{(k+1)} = \sigma(\widetilde{A}H^{(k)}W^{(k)}). \tag{16}$$

Intuitively, we consider combining a simple graph convolution defined via the polynomial $I + A$ with nonlinearities and trainable weight matrices, then the formula above can be rewritten as

$$H^{(k+1)} = \sigma\left(AH^{(k)}W^{(k)}_{\text{neigh}} + H^{(k)}W^{(k)}_{\text{self}}\right). \tag{17}$$

That is to say, a simple graph convolution based on $I + A$ is equivalent to aggregating information from the neighboring nodes defined by the adjacency matrix and the node itself.

Appendix B
Naïve Bayes Classifier

A. Naïve Bayes

Naïve Bayes is a classification method based on Bayes' theorem and conditional independence assumption.

Assume that the input space $\mathcal{X} \in \mathbb{R}^n$ is a set of n-dimensional vectors, the output space is the class label set $\mathcal{Y} = \{c_1, c_2, \ldots, c_K\}$. The input is the feature vector $x \in \mathcal{X}$ and the output is the class label $y \in \mathcal{Y}$. Accordingly, X and Y are random variables defined in $\mathcal{X}$ and $\mathcal{Y}$, respectively. $P(X, Y)$ is the joint probability distribution of X and Y.

Naïve Bayes learns joint probability distribution through the training data. Specifically, it learns the prior and conditional probability distribution. The conditional probability distribution is denoted as

$$P(X=x|Y=c_k) = P(X^{(1)}=x^{(1)}, \ldots, X^{(n)}=x^{(n)}|Y=c_k), \quad k = 1, 2, \ldots, K. \tag{18}$$

The conditional independence assumption can be formulated as

$$\begin{aligned} P(X=x|Y=c_k) &= P(X^{(1)}=x^{(1)}, \ldots, X^{(n)}=x^{(n)}|Y=c_k) \\ &= \prod_{i=1}^{n} P(X^{(i)} = x^{(i)}|Y = c_k). \end{aligned} \tag{19}$$

Here, we directly give the conclusion that naïve Bayes take the class with the highest posterior probability as the output. Combined with the conditional independence assumption, it can be expressed as follows:

$$y = \arg\max_{c_k} P(Y = c_k) \prod_{i=1}^{n} P(X^{(i)} = x^{(i)}|Y = c_k). \tag{20}$$

B. Maximum A Posteriori Estimation

Naïve Bayes classifies the instance into the class with the highest posterior probability. This is equivalent to expectation risk minimization. Assume that it is zero-one loss function as

$$L(Y, f(X)) = \begin{cases} 1, & Y \neq f(X) \\ 0, & Y = f(X) \end{cases} \tag{21}$$

where $f(X)$ is the classification decision function. Here, the expectation risk function is

$$R_{\text{exp}}(f) = E[L(Y, f(X))]. \tag{22}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:44:10 UTC from IEEE Xplore. Restrictions apply.

The expectation is for the joint probability distribution $P(X, Y)$, so we take the conditional expectation

$$R_{\exp}(f) = E_X \sum_{k=1}^{K} [L(c_k, f(X))] P(c_k|X). \tag{23}$$

To minimize the expectation risk, we just need to minimize the value. Therefore, the result can be formulated as

$$\begin{aligned} f(x) &= \arg\min_{y\in\mathcal{Y}} \sum_{k=1}^{K} L(c_k, y) P(c_k|X = x) \\ &= \arg\min_{y\in\mathcal{Y}} \sum_{k=1}^{K} P(y \neq c_k|X = x) \\ &= \arg\min_{y\in\mathcal{Y}} (1 - P(y = c_k|X = x)) \\ &= \arg\max_{y\in\mathcal{Y}} P(y = c_k|X = x). \end{aligned} \tag{24}$$

In this way, it leads to posterior probability maximization according to expectation risk minimization

$$f(x) = \arg\max_{c_k} P(c_k|X = x). \tag{25}$$

which is exactly the principle of naïve Bayes.

REFERENCES

[1] Z. Zhang, Q. Liu, and Y. Wang, "Road extraction by deep residual U-Net," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 5, pp. 749–753, May 2018.

[2] A. S. Goodman and M. Hastak, *Infrastructure Planning Handbook: Planning, Engineering, and Economics*. Reston, VA, USA: ASCE, 2006.

[3] Q. Zhang and K. C. Seto, "Mapping urbanization dynamics at regional and global scales using multi-temporal DMSP/OLS nighttime light data," *Remote Sens. Environ.*, vol. 115, no. 9, pp. 2320–2329, Sep. 2011.

[4] Z. Zheng, Y. Zhong, J. Wang, and A. Ma, "Foreground-aware relation network for geospatial object segmentation in high spatial resolution remote sensing imagery," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 4096–4105.

[5] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2017.

[6] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," 2017, *arXiv:1706.05587*. [Online]. Available: http://arxiv.org/abs/1706.05587

[7] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder-decoder with atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 801–818.

[8] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2881–2890.

[9] X. Wang, R. Girshick, A. Gupta, and K. He, "Non-local neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7794–7803.

[10] Z. Huang, X. Wang, L. Huang, C. Huang, Y. Wei, and W. Liu, "CCNet: Criss-cross attention for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 603–612.

[11] J. Fu *et al.*, "Dual attention network for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3146–3154.

[12] S. Huang *et al.*, "ORDNet: Capturing omni-range dependencies for scene parsing," *IEEE Trans. Image Process.*, vol. 29, pp. 8251–8263, 2020.

[13] X. Li *et al.*, "PointFlow: Flowing semantics through points for aerial image segmentation," 2021, *arXiv:2103.06564*. [Online]. Available: http://arxiv.org/abs/2103.06564

[14] R. Niu, X. Sun, Y. Tian, W. Diao, K. Chen, and K. Fu, "Hybrid multiple attention network for semantic segmentation in aerial images," *IEEE Trans. Geosci. Remote Sens.*, early access, Mar. 24, 2021, doi: 10.1109/TGRS.2021.3065112.

[15] X. Liang, Z. Hu, H. Zhang, L. Lin, and E. P. Xing, "Symbolic graph reasoning meets convolutions," in *Proc. 32nd Int. Conf. Neural Inf. Process. Syst.*, 2018, pp. 1858–1868.

[16] Y. Li and A. Gupta, "Beyond grids: Learning graph representations for visual recognition," in *Proc. 32nd Int. Conf. Neural Inf. Process. Syst.*, 2018, pp. 9245–9255.

[17] L. Zhang, D. Xu, A. Arnab, and P. H. S. Torr, "Dynamic graph message passing networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 3726–3735.

[18] Y. Chen, M. Rohrbach, Z. Yan, Y. Shuicheng, J. Feng, and Y. Kalantidis, "Graph-based global reasoning networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 433–442.

[19] C. Yu, Y. Liu, C. Gao, C. Shen, and N. Sang, "Representative graph neural network," in *Proc. Eur. Conf. Comput. Vis.* Berlin, Germany: Springer, 2020, pp. 379–396.

[20] H. Hu, D. Ji, W. Gan, S. Bai, W. Wu, and J. Yan, "Class-wise dynamic graph convolution for semantic segmentation," 2020, *arXiv:2007.09690*. [Online]. Available: http://arxiv.org/abs/2007.09690

[21] T.-Y. Lin, A. RoyChowdhury, and S. Maji, "Bilinear CNN models for fine-grained visual recognition," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1449–1457.

[22] Y. Bengio, A. Courville, and P. Vincent, "Representation learning: A review and new perspectives," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 8, pp. 1798–1828, Aug. 2013.

[23] I. Higgins *et al.*, "β-VAE: Learning basic visual concepts with a constrained variational framework," in *Proc. 5th Int. Conf. Learn. Represent. (ICLR)*, Toulon, France, Apr. 2017.

[24] D. P. Kingma and M. Welling, "Auto-encoding variational Bayes," 2013, *arXiv:1312.6114*. [Online]. Available: http://arxiv.org/abs/1312.6114

[25] S. Waqas Zamir *et al.*, "iSAID: A large-scale dataset for instance segmentation in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR) Workshops*, Jun. 2019, pp. 28–37.

[26] *ISPRS 2D Semantic Labeling Contest Vaihingen*. Accessed: Aug. 23, 2020. [Online]. Available: http://www2.isprs.org/commissions/comm3/wg4/2d-sem-label-vaihingen.html

[27] M. Cordts *et al.*, "The cityscapes dataset for semantic urban scene understanding," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 3213–3223.

[28] V. Badrinarayanan, A. Kendall, and R. Cipolla, "SegNet: A deep convolutional encoder-decoder architecture for image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Dec. 2017.

[29] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "Semantic image segmentation with deep convolutional nets and fully connected CRFs," 2014, *arXiv:1412.7062*. [Online]. Available: http://arxiv.org/abs/1412.7062

[30] G. Lin, A. Milan, C. Shen, and I. Reid, "RefineNet: Multi-path refinement networks for high-resolution semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 1925–1934.

[31] M. Yang, K. Yu, C. Zhang, Z. Li, and K. Yang, "DenseASPP for semantic segmentation in street scenes," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3684–3692.

[32] H. Li, P. Xiong, H. Fan, and J. Sun, "DFANet: Deep feature aggregation for real-time semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 9522–9531.

[33] H. Zhang *et al.*, "Context encoding for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7151–7160.

[34] T. Takikawa, D. Acuna, V. Jampani, and S. Fidler, "Gated-SCNN: Gated shape CNNs for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 5229–5238.

[35] Z. Huang, Y. Wei, X. Wang, H. Shi, W. Liu, and T. S. Huang, "AlignSeg: Feature-aligned segmentation networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, early access, Mar. 1, 2021, doi: 10.1109/TPAMI.2021.3062772.

[36] Y. Feng, X. Sun, W. Diao, J. Li, and X. Gao, "Double similarity distillation for semantic image segmentation," *IEEE Trans. Image Process.*, vol. 30, pp. 5363–5376, 2021.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:44:10 UTC from IEEE Xplore. Restrictions apply.

[37] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.

[38] A. Vaswani *et al.*, "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 5998–6008.

[39] M. Yin *et al.*, "Disentangled non-local neural networks," in *Proc. Eur. Conf. Comput. Vis.* Berlin, Germany: Springer, 2020, pp. 191–207.

[40] Z. Zhu, M. Xu, S. Bai, T. Huang, and X. Bai, "Asymmetric non-local neural networks for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 593–602.

[41] X. Li, Z. Zhong, J. Wu, Y. Yang, Z. Lin, and H. Liu, "Expectation-maximization attention networks for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9167–9176.

[42] Y. Cao, J. Xu, S. Lin, F. Wei, and H. Hu, "GCNet: Non-local networks meet squeeze-excitation networks and beyond," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshop (ICCVW)*, Oct. 2019, pp. 1971–1980.

[43] Y. Yuan, X. Chen, X. Chen, and J. Wang, "Segmentation transformer: Object-contextual representations for semantic segmentation," 2019, *arXiv:1909.11065*. [Online]. Available: http://arxiv.org/abs/1909.11065

[44] Y. Chen, Y. Kalantidis, J. Li, S. Yan, and J. Feng, "A^2-nets: Double attention networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2018, pp. 352–361.

[45] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 3–19.

[46] X. Li *et al.*, "Semantic flow for fast and accurate scene parsing," in *Proc. Eur. Conf. Comput. Vis.* Berlin, Germany: Springer, 2020, pp. 775–793.

[47] X. Li *et al.*, "Improving semantic segmentation via decoupled body and edge supervision," 2020, *arXiv:2007.10035*. [Online]. Available: http://arxiv.org/abs/2007.10035

[48] M. Jaderberg, K. Simonyan, A. Zisserman, and K. Kavukcuoglu, "Spatial transformer networks," 2015, *arXiv:1506.02025*. [Online]. Available: http://arxiv.org/abs/1506.02025

[49] D. Mazzini, "Guided upsampling network for real-time semantic segmentation," 2018, *arXiv:1807.07466*. [Online]. Available: http://arxiv.org/abs/1807.07466

[50] J. Dai *et al.*, "Deformable convolutional networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 764–773.

[51] A. Dosovitskiy *et al.*, "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*. [Online]. Available: http://arxiv.org/abs/2010.11929

[52] Z. Liu *et al.*, "Swin transformer: Hierarchical vision transformer using shifted Windows," 2021, *arXiv:2103.14030*. [Online]. Available: http://arxiv.org/abs/2103.14030

[53] S. Zheng *et al.*, "Rethinking semantic segmentation from a sequence-to-sequence perspective with transformers," 2021, *arXiv:2012.15840*. [Online]. Available: http://arxiv.org/abs/2012.15840

[54] L. Mou, Y. Hua, and X. X. Zhu, "Relation matters: Relational context-aware fully convolutional network for semantic segmentation of high-resolution aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 7557–7569, Nov. 2020.

[55] G. Deng, Z. Wu, C. Wang, M. Xu, and Y. Zhong, "CCANet: Class-constraint coarse-to-fine attentional deep network for subdecimeter aerial image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, early access, Feb. 12, 2021, doi: 10.1109/TGRS.2021.3055950.

[56] A. Li, L. Jiao, H. Zhu, L. Li, and F. Liu, "Multitask semantic boundary awareness network for remote sensing image segmentation," *IEEE Trans. Geosci. Remote Sens.*, early access, Feb. 2, 2021, doi: 10.1109/TGRS.2021.3050885.

[57] X. Sun, A. Shi, H. Huang, and H. Mayer, "Bas4Net: Boundary-aware semi-supervised semantic segmentation network for very high resolution remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 5398–5413, Sep. 2020.

[58] L. Mi and Z. Chen, "Superpixel-enhanced deep neural forest for remote sensing image semantic segmentation," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 140–152, Jan. 2020.

[59] C. Peng, K. Zhang, Y. Ma, and J. Ma, "Cross fusion net: A fast semantic segmentation network for small-scale semantic information capturing in aerial scenes," *IEEE Trans. Geosci. Remote Sens.*, early access, Jan. 29, 2021, doi: 10.1109/TGRS.2021.3053062.

[60] Y. Feng *et al.*, "NPALoss: Neighboring pixel affinity loss for semantic segmentation in high-resolution aerial imagery," *ISPRS Ann. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 2, no. 2, pp. 475–482, 2020.

[61] A. Ma, J. Wang, Y. Zhong, and Z. Zheng, "FactSeg: Foreground activation-driven small object semantic segmentation in large-scale remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, early access, Jul. 27, 2021, doi: 10.1109/TGRS.2021.3097148.

[62] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," 2016, *arXiv:1609.02907*. [Online]. Available: http://arxiv.org/abs/1609.02907

[63] Y. Bi, A. Chadha, A. Abbas, E. Bourtsoulatze, and Y. Andreopoulos, "Graph-based object classification for neuromorphic vision sensing," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 491–501.

[64] X. Wang and A. Gupta, "Videos as space-time region graphs," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 399–417.

[65] Z. Zhang *et al.*, "PPGNet: Learning point-pair graph for line segment detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 7105–7114.

[66] H. Xu, C. Jiang, X. Liang, and Z. Li, "Spatial-aware graph relation network for large-scale object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 9298–9307.

[67] O. Ulutan, A. S. M. Iftekhar, and B. S. Manjunath, "VSGNet: Spatial attention network for detecting human object interactions using graph convolutions," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 13617–13626.

[68] X. Li, Y. Yang, Q. Zhao, T. Shen, Z. Lin, and H. Liu, "Spatial pyramid based graph reasoning for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 8950–8959.

[69] X. Chen *et al.*, "Variational lossy autoencoder," 2016, *arXiv:1611.02731*. [Online]. Available: http://arxiv.org/abs/1611.02731

[70] A. A. Alemi, I. Fischer, J. V. Dillon, and K. Murphy, "Deep variational information bottleneck," 2016, *arXiv:1612.00410*. [Online]. Available: http://arxiv.org/abs/1612.00410

[71] H. Kim and A. Mnih, "Disentangling by factorising," in *Proc. Int. Conf. Mach. Learn.*, 2018, pp. 2649–2658.

[72] D. J. Rezende, S. Mohamed, and D. Wierstra, "Stochastic backpropagation and approximate inference in deep generative models," in *Proc. Int. Conf. Mach. Learn.*, 2014, pp. 1278–1286.

[73] N. Tishby and N. Zaslavsky, "Deep learning and the information bottleneck principle," in *Proc. IEEE Inf. Theory Workshop (ITW)*, Apr. 2015, pp. 1–5.

[74] N. Tishby, F. C. Pereira, and W. Bialek, "The information bottleneck method," 2000, *arXiv:physics/0004057*. [Online]. Available: https://arxiv.org/abs/physics/0004057

[75] R. T. Q. Chen, X. Li, R. Grosse, and D. Duvenaud, "Isolating sources of disentanglement in variational autoencoders," 2018, *arXiv:1802.04942*. [Online]. Available: http://arxiv.org/abs/1802.04942

[76] C. P. Burgess *et al.*, "Understanding disentangling in β-VAE," 2018, *arXiv:1804.03599*. [Online]. Available: https://arxiv.org/abs/1804.03599

[77] Y.-C. Liu, Y.-Y. Yeh, T.-C. Fu, S.-D. Wang, W.-C. Chiu, and Y.-C.-F. Wang, "Detach and adapt: Learning cross-domain disentangled deep representation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8867–8876.

[78] J. Ma, C. Zhou, P. Cui, H. Yang, and W. Zhu, "Learning disentangled representations for recommendation," 2019, *arXiv:1910.14238*. [Online]. Available: http://arxiv.org/abs/1910.14238

[79] J. Ma, P. Cui, K. Kuang, X. Wang, and W. Zhu, "Disentangled graph convolutional networks," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 4212–4221.

[80] X. Li *et al.*, "Towards efficient scene understanding via squeeze reasoning," 2020, *arXiv:2011.03308*. [Online]. Available: http://arxiv.org/abs/2011.03308

[81] C. Yu, X. Zhao, Q. Zheng, P. Zhang, and X. You, "Hierarchical bilinear pooling for fine-grained visual recognition," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 574–589.

[82] J.-H. Kim, K.-W. On, W. Lim, J. Kim, J.-W. Ha, and B.-T. Zhang, "Hadamard product for low-rank bilinear pooling," 2016, *arXiv:1610.04325*. [Online]. Available: http://arxiv.org/abs/1610.04325

[83] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.

[84] Y. Yuan, J. Xie, X. Chen, and J. Wang, "SegFix: Model-agnostic boundary refinement for segmentation," in *Proc. Eur. Conf. Comput. Vis.* Berlin, Germany: Springer, 2020, pp. 489–506.

[85] S. Choi, J. T. Kim, and J. Choo, "Cars can't fly up in the sky: Improving urban-scene segmentation via height-driven attention networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2020, pp. 9373–9383.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:44:10 UTC from IEEE Xplore. Restrictions apply.

[86] A. Dosovitskiy *et al.*, "FlowNet: Learning optical flow with convolutional networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 2758–2766.
[87] E. Ilg, N. Mayer, T. Saikia, M. Keuper, A. Dosovitskiy, and T. Brox, "FlowNet 2.0: Evolution of optical flow estimation with deep networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2462–2470.
[88] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollar, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.
[89] F. Milletari, N. Navab, and S.-A. Ahmadi, "V-Net: Fully convolutional neural networks for volumetric medical image segmentation," in *Proc. 4th Int. Conf. 3D Vis. (3DV)*, Oct. 2016, pp. 565–571.
[90] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[91] O. Russakovsky *et al.*, "ImageNet large scale visual recognition challenge," *Int. J. Comput. Vis.*, vol. 115, no. 3, pp. 211–252, Dec. 2015.
[92] S. R. Bulo, L. Porzi, and P. Kontschieder, "In-place activated BatchNorm for memory-optimized training of DNNs," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 5639–5647.
[93] H. Liu *et al.*, "An end-to-end network for panoptic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 6172–6181.
[94] T. Xiao, Y. Liu, B. Zhou, Y. Jiang, and J. Sun, "Unified perceptual parsing for scene understanding," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 418–434.
[95] J. Wang *et al.*, "Deep high-resolution representation learning for visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 10, pp. 3349–3364, Oct. 2021.
[96] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, "Graph attention networks," 2017, *arXiv:1710.10903*. [Online]. Available: http://arxiv.org/abs/1710.10903
[97] J. Yu and M. Blaschko, "Learning submodular losses with the Lovász Hinge," in *Proc. Int. Conf. Mach. Learn.*, 2015, pp. 1623–1631.
[98] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.* Berlin, Germany: Springer, 2015, pp. 234–241.
[99] D. Marcos, M. Volpi, B. Kellenberger, and D. Tuia, "Land cover mapping at very high resolution with rotation equivariant CNNs: Towards small yet accurate models," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 96–107, Nov. 2018.
[100] N. Audebert, B. Le Saux, and S. Lefèvre, "Beyond RGB: Very high resolution urban remote sensing with multimodal deep networks," *ISPRS J. Photogramm. Remote Sens.*, vol. 140, pp. 20–32, Jun. 2018.
[101] D. Marmanis, K. Schindler, J. D. Wegner, S. Galliani, M. Datcu, and U. Stilla, "Classification with an edge: Improving semantic image segmentation with boundary detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 135, pp. 158–172, Jan. 2018.
[102] Y. Liu, B. Fan, L. Wang, J. Bai, S. Xiang, and C. Pan, "Semantic labeling in very high resolution images via a self-cascaded convolutional neural network," *ISPRS J. Photogram. Remote Sens.*, vol. 145, pp. 78–95, Nov. 2018.
[103] Y. Sun, Y. Tian, and Y. Xu, "Problems of encoder-decoder frameworks for high-resolution remote sensing image segmentation: Structural stereotype and insufficient learning," *Neurocomputing*, vol. 330, pp. 297–304, Feb. 2019.
[104] H. Zhao *et al.*, "PSANet: Point-wise spatial attention network for scene parsing," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 267–283.
[105] C. Yu, J. Wang, C. Gao, G. Yu, C. Shen, and N. Sang, "Context prior for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 12416–12425.

Ruigang Niu (Graduate Student Member, IEEE) received the B.Sc. degree from Northwestern Polytechnical University, Xi'an, China, in 2018. He is currently pursuing the Ph.D. degree with the University of Chinese Academy of Sciences, Beijing, China, and the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing.

His research interests include computer vision, pattern recognition, and remote sensing image processing, especially on semantic segmentation.

Xian Sun (Senior Member, IEEE) received the B.Sc. degree from the Beijing University of Aeronautics and Astronautics, Beijing, China, in 2004, and the M.Sc. and Ph.D. degrees from the Institute of Electronics, Chinese Academy of Sciences, Beijing, in 2009.

He is currently a Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences. His research interests include computer vision, geospatial data mining, and remote sensing image understanding.

Yu Tian received the B.Sc. degree from Dalian Maritime University, Dalian, China, in 2018. He is currently pursuing the Ph.D. degree with the University of Chinese Academy of Sciences, Beijing, China, and the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing.

His research interests include computer vision, pattern recognition, and remote sensing image processing, especially on instance segmentation.

Wenhui Diao (Member, IEEE) received the B.Sc. degree from Xidian University, Xi'an, China, in 2011, and the M.Sc. and Ph.D. degrees from the Institute of Electronics, Chinese Academy of Sciences, Beijing, China, in 2016.

He is currently an Assistant Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences. His research interests include computer vision and remote sensing image analysis.

Yingchao Feng (Graduate Student Member, IEEE) received the bachelor's degree from Xidian University, Xi'an, China, in 2017. He is currently pursuing the Ph.D. degree with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China.

His research interests include computer vision and deep learning, especially on object detection, semantic segmentation, and remote sensing.

Kun Fu (Member, IEEE) received the B.Sc., M.Sc., and Ph.D. degrees from the National University of Defense Technology, Changsha, China, in 1995, 1999, and 2002, respectively.

He is currently a Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China. His research interests include computer vision, remote sensing image understanding, geospatial data mining, and visualization.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:44:10 UTC from IEEE Xplore. Restrictions apply.